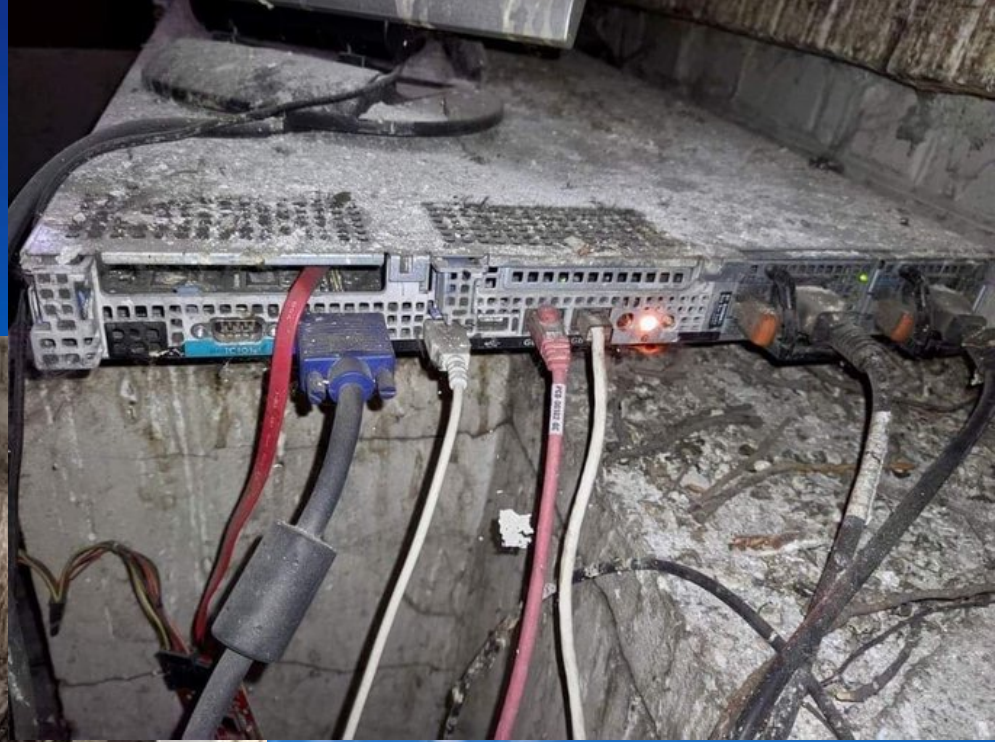


Broadcast Technology UG1

DVB, Compression, TS, PS

DVB
MPEG compression
PS,TS, Streaming
IP Workflows

**“The Cloud is just
somebody else’s
computer”**



**“Your data is safe
in our cloud”**



This Session:

Networks

- TDM
- Circuit Switching
- Packet Switching

Layers of Networks

- Telephone, Internet, WWW

MPEG Systems

- Multiplexes
- Program Streams
- Transport Streams
- Statistical Multiplexing

DVB

Streaming

- HTTP
- ABR, DASH

SDI and IP

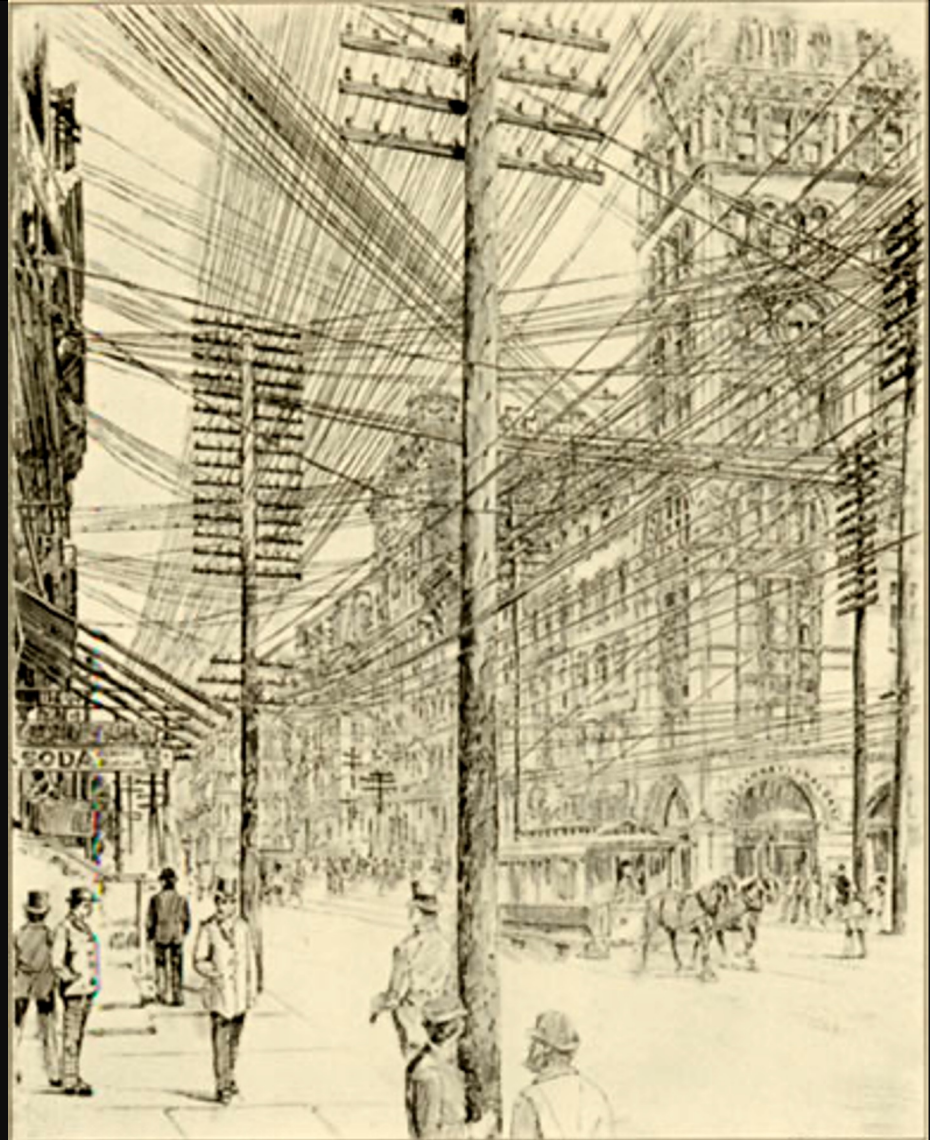
SMPTE 2022, 2110

Compression

- General
- Audio
- Video

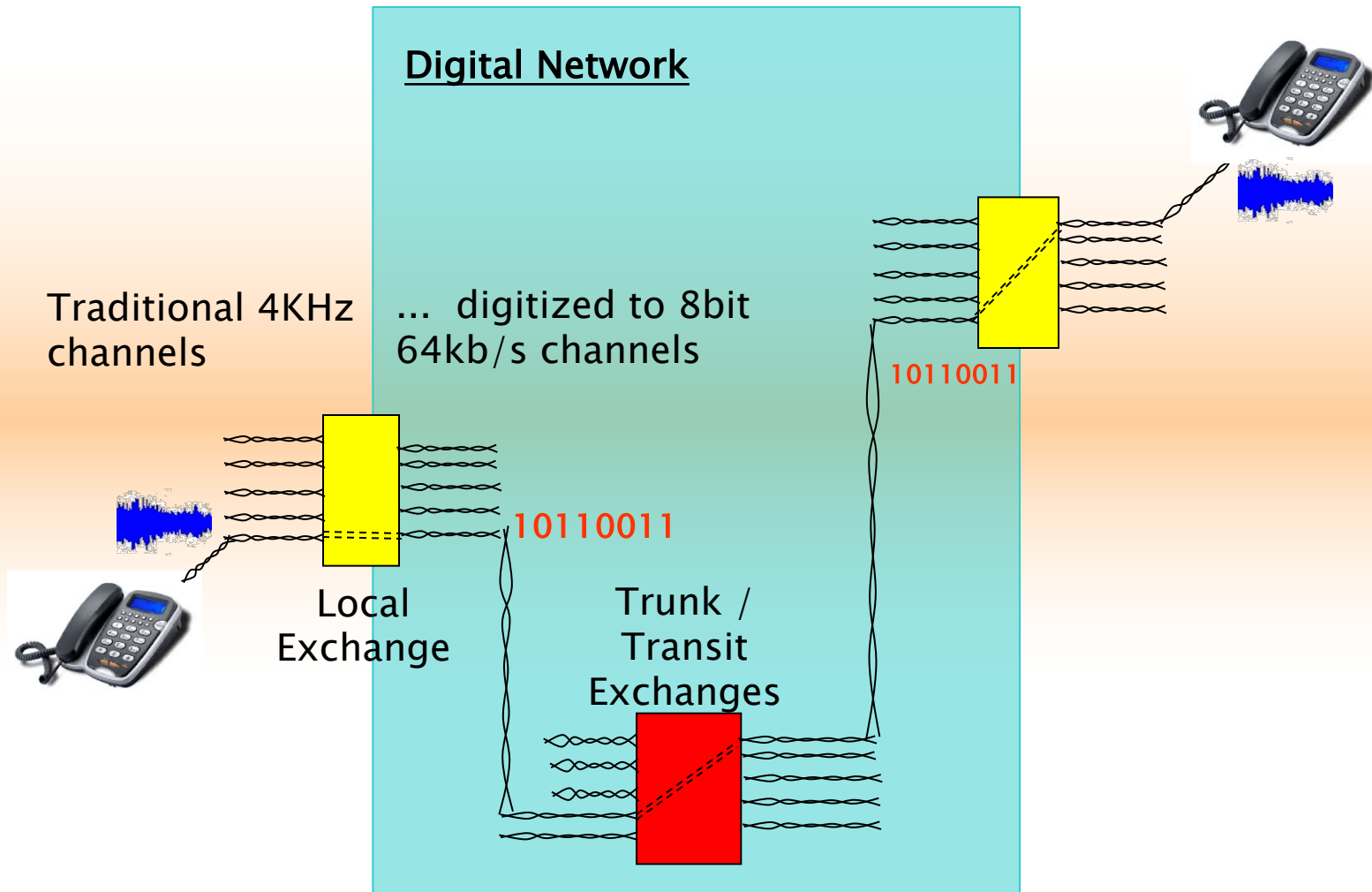
From Circuits

...

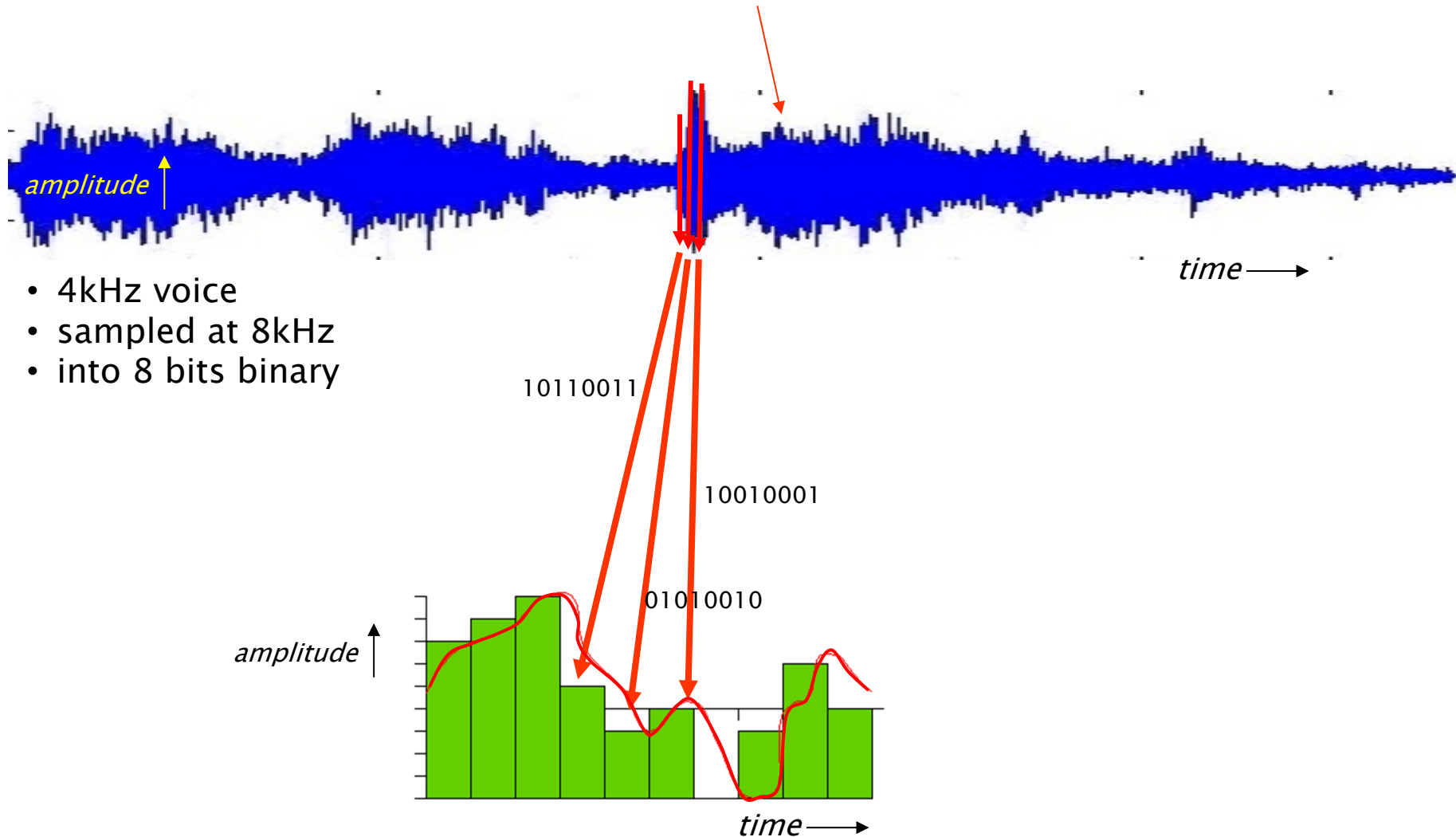


... to Packets

Traditional PSTN – Digitisation of the Network

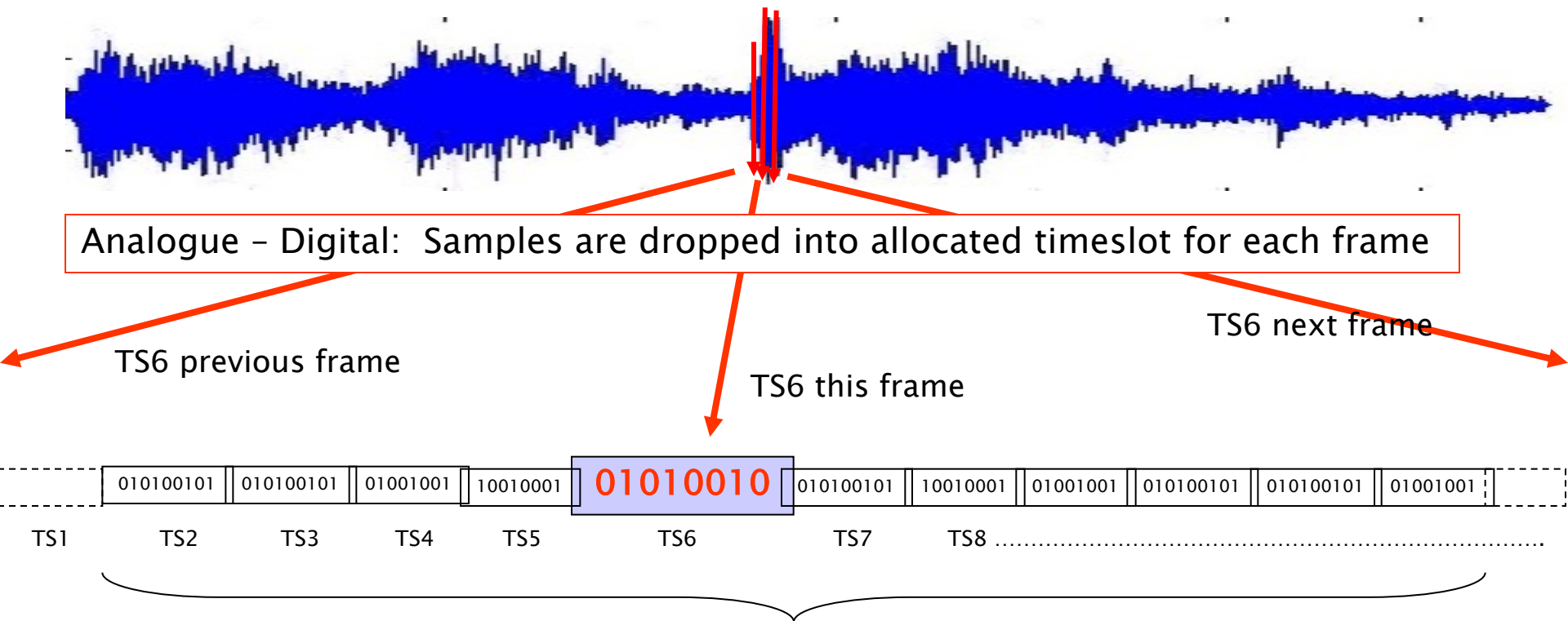


Time Division Multiplex



Time Division Multiplex

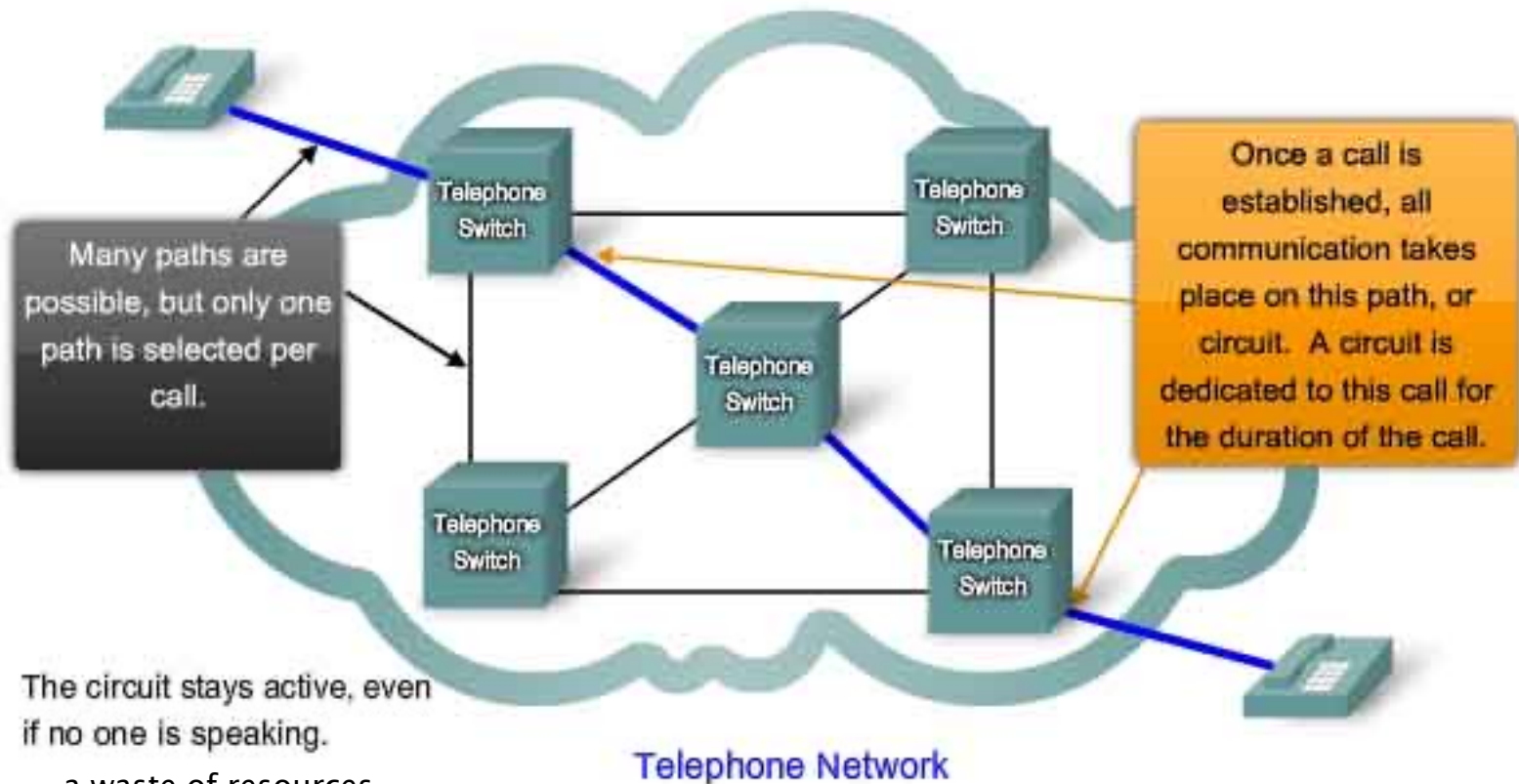
Many phone calls at the same time allocated to other slots in the multiplex



30 channels of speech into a time-divided-multiplex

Circuit Switching in a Network

- economy of circuits ✗
- resilience / survivability -
- latency ✓



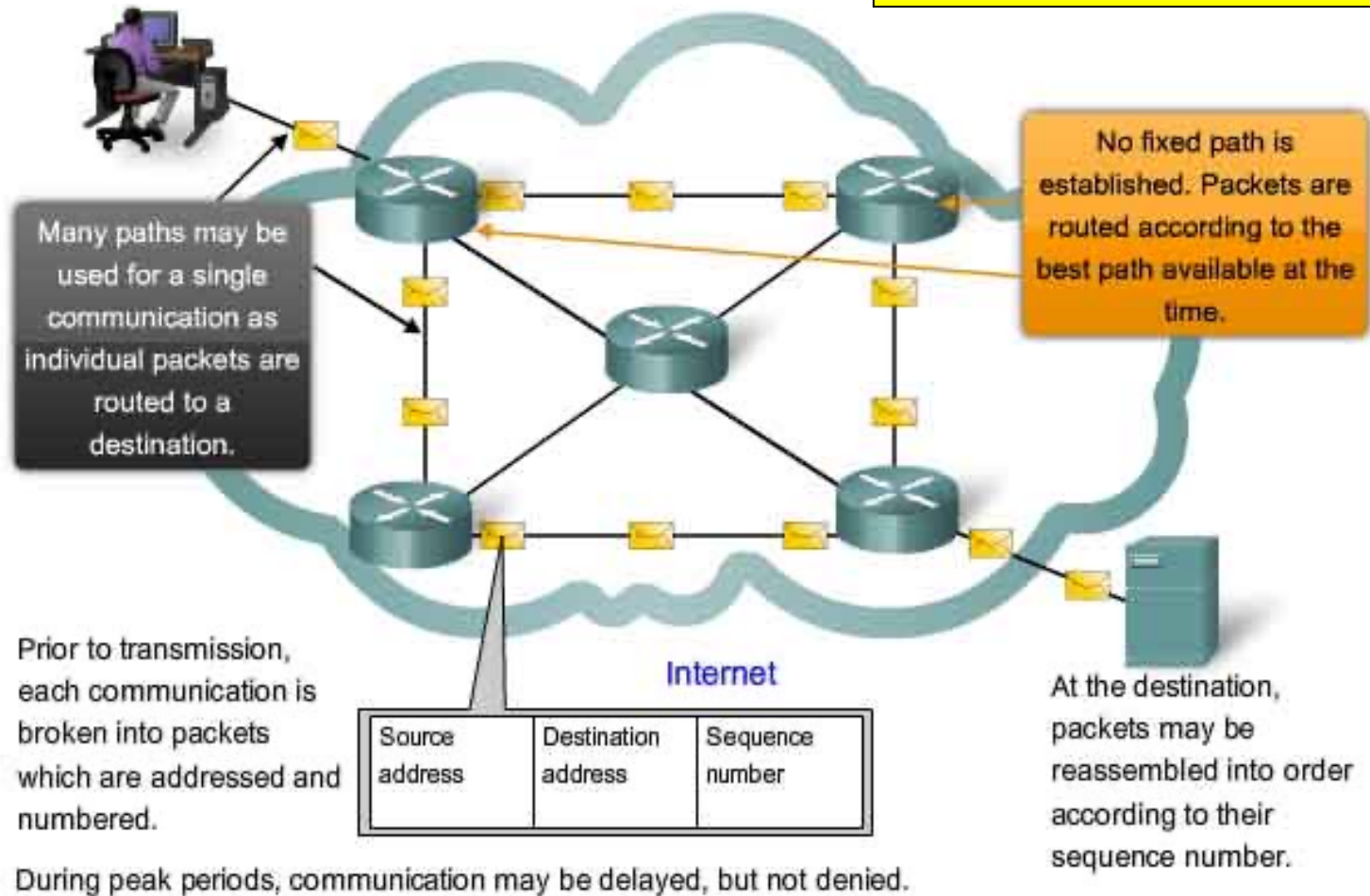
The circuit stays active, even if no one is speaking.

... a waste of resources when there is no data to transmit

<http://www.highteck.net/EN/Basic/Internetworking.html>

Packet Switching in a Network

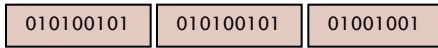
- economy of circuits ✓
- resilience / survivability ✓
- latency ✗



<http://www.highteck.net/EN/Basic/Internetworking.html>

Packet Switching

digitised voice



digital data

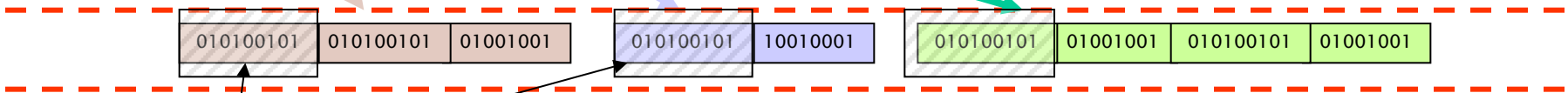


digital media

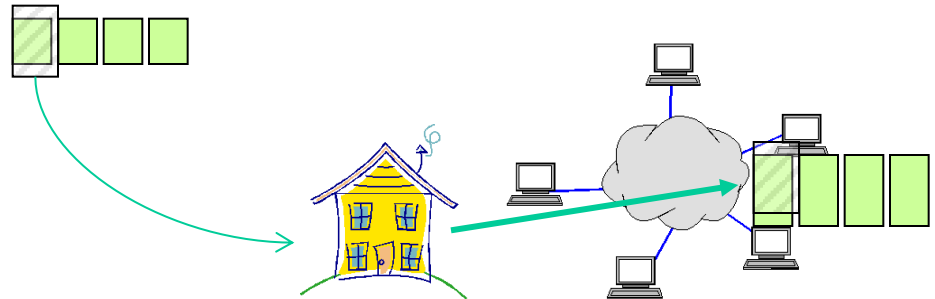


- all traffic – voice, video, data is split into *packets*
- packets are dropped into a *bitstream* as/when available
... “*statistically multiplexed*”
... the bitstream may not always have enough capacity
for all the traffic *competing* for access

internet network link

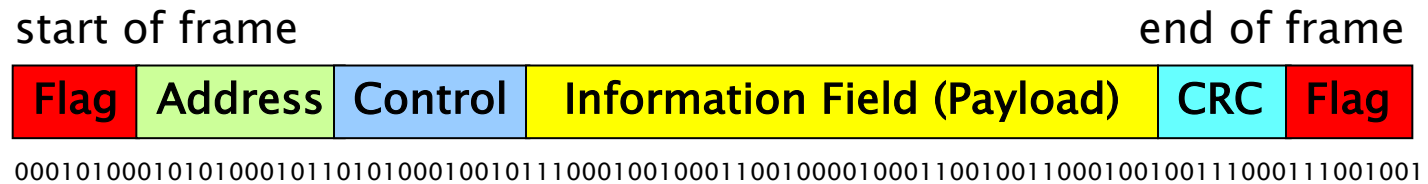


address field – packets
routed through network



House / business ethernet (or wireless) LAN

Message in Packets

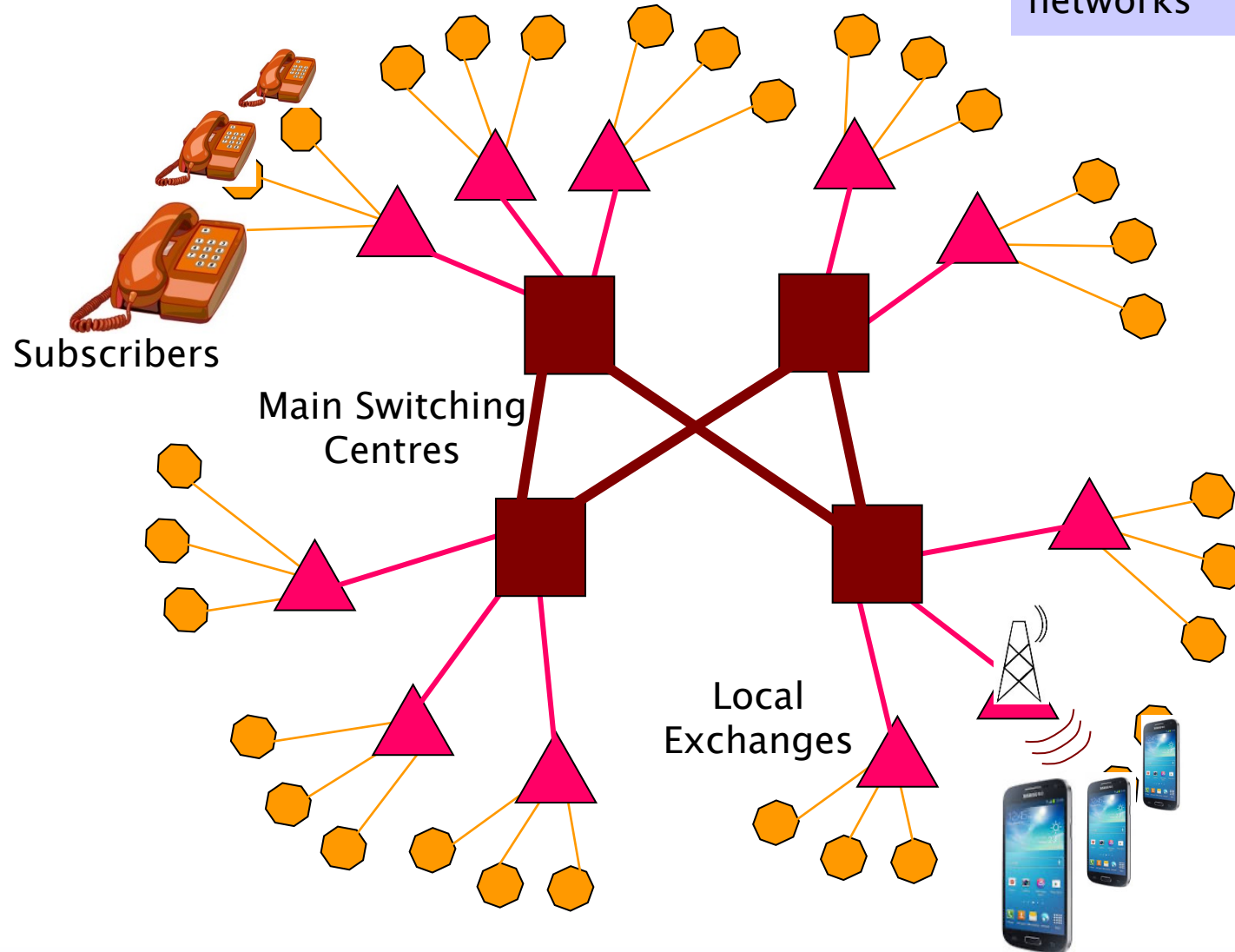


Layers of Networks

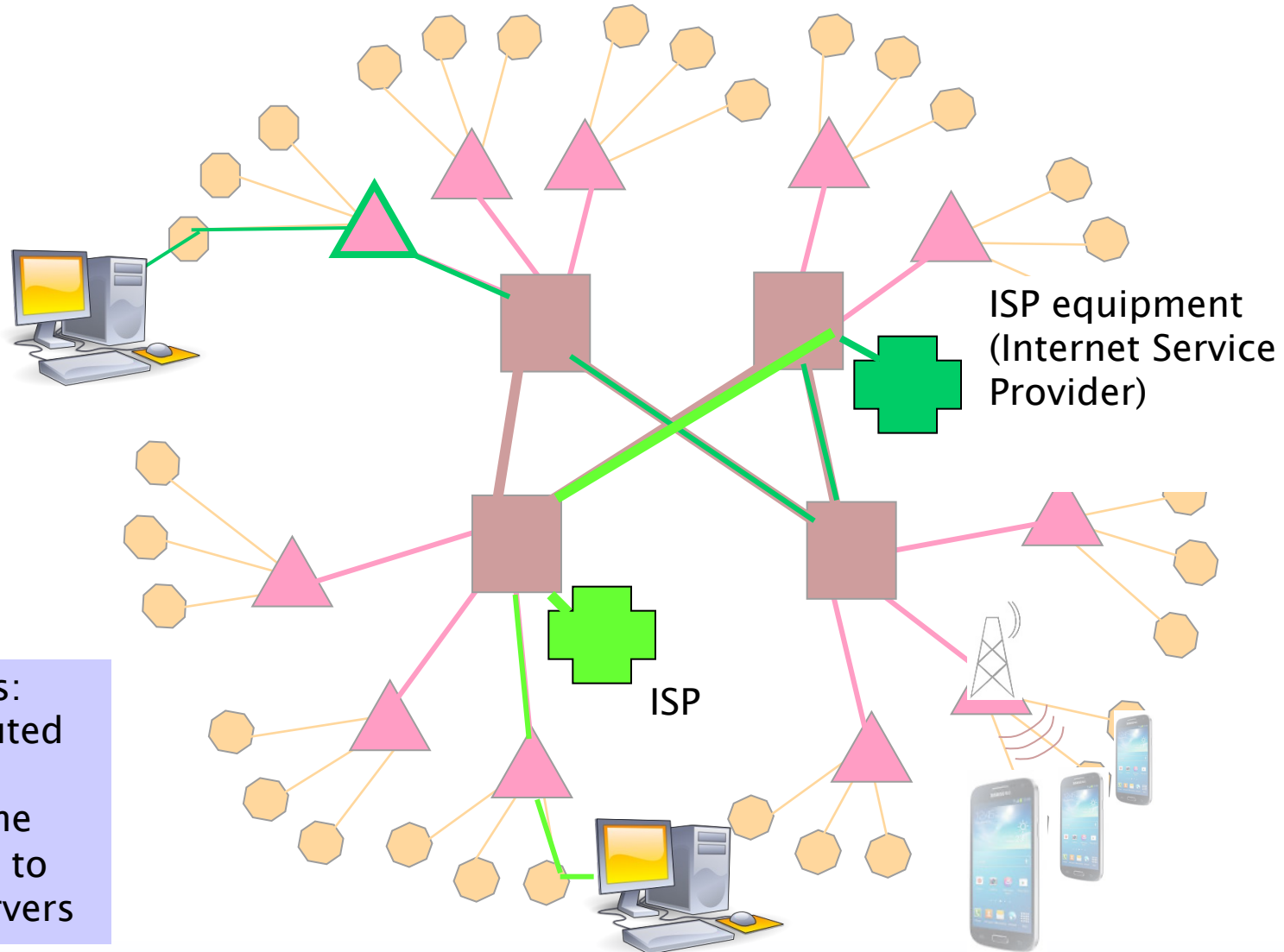
- The Carrier Network
- Ethernet (Data Link)
- Internet
 - (+ The Web)
- Protocols
- Routing

Telecomm Network

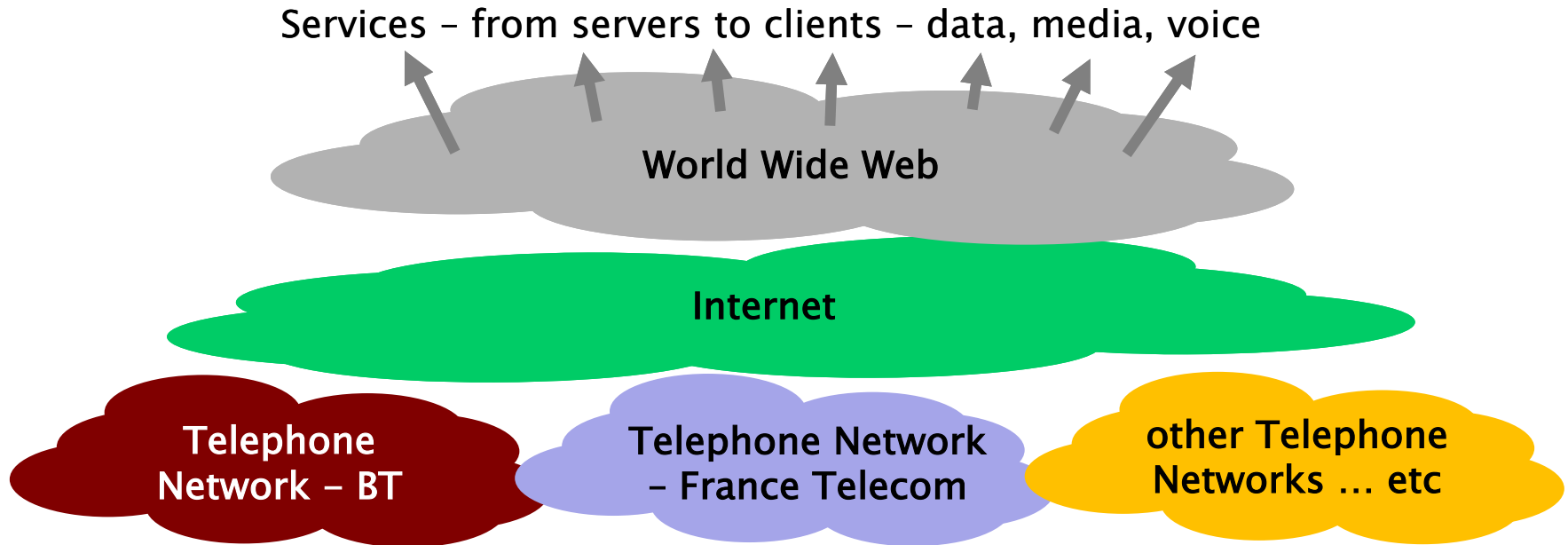
Internet Protocol – IP –
to carry data, voice and
media over telephone
networks



Telephone Network – Internet Overlay



Layers of Networks



Internet

- US, mid '60's ARPANET; DOD Universities
- data (voice, video, data) is sent in packets
- packets are routed around the network individually
- “survivable network”

MPEG Systems

MPEG family

Overviews

- MPEG-1, MPEG-2, MPEG-2 Audio, MPEG4

Audio Compression

- Compression Usability, Latency
- Encoding Architecture
- Perceptual Coding: Psychoacoustic Models
- Transform Domain Methods, Decoding
- Coding, Bit Allocation, Frames

AAC – Advanced Audio Coding

- MPEG-4 AAC, Low Delay Coder, Error Resilience,
- AAC compared to mp3

Other Formats

Variable Bit Rate

MPEG

Motion Picture Experts Group

MPEG 1

audio and video coding – parts 1, 2, 3 (layers 1, 2, 3), 4, 5,

MPEG 2

audio and video coding – parts 1, 2, 3 (layers 1, 2, 3), ...

MPEG 3 – originally for HDTV – abandoned

MPEG 4

audio and video coding – parts 1 – 22

MPEG 7

metadata, description

MPEG 21

framework for multimedia content delivery

- XML-based; to communicate machine-readable license information
- open framework for multimedia applications
- "Rights Expression Language"

MPEG–1

Audio and video coding

Parts

- 1 – Synchronization and multiplexing of video and audio
 - transport stream
 - 2 – Compression codec for non-interlaced video signals
 - 3 – **Compression codec for perceptual coding of audio signals**
 - three layers of audio coding options (mp, mp2, mp3)
 - 4 – Procedures for testing conformance
 - 5 – Reference software
- video CD format – quality ~ VCR
 - originally 1.5M bit/second data rates and 352x240 resolution
 - later up to 4M bit/second
 - the most compatible format in the MPEG family
 - only progressive scan video (... so need for MPEG–2)

MPEG-2

(1994) Audio and video coding

- for broadcast – DVB (satellite, cable, DTT, etc), also DVD (with modifications)

9 Parts, including –

1 – Transport Streams – TS

- to carry digital video and audio in (unreliable) broadcast applications
- Program streams carry audio and/or video data + time stamp data

2 – Video

- similar to MPEG-1
- support for interlaced video
- optimised > 3 Mbit/s; *not* for low bit-rates (less than 1 Mbit/s)
- backwards compatible to MPEG1
- enhancements – used (to date) in most HDTV transmission systems

3 – Audio

- layers 1, 2, 3

4, 5 – as MPEG-1

7 – Enhanced audio

MPEG-2 Audio

Part 3

- new audio encoding (see later...)
- low bitrate encoding with halved sampling rate
 - MPEG-1 Layer 1 / 2 / 3 LSF – low sampling frequency
- multichannel encoding with up to 5.1 channels
 - ... backwards compatible –
MPEG-1 audio decoders can decode stereo from MPEG-2 multi-chan audio

MPEG-2 AAC – Part 7

- make better use of available bandwidth
 - non-backwards-compatible

Streaming Container Formats

“Over the air” /satellite/cable streaming

- using transport streams (TS)
 - c.f. Program Streams

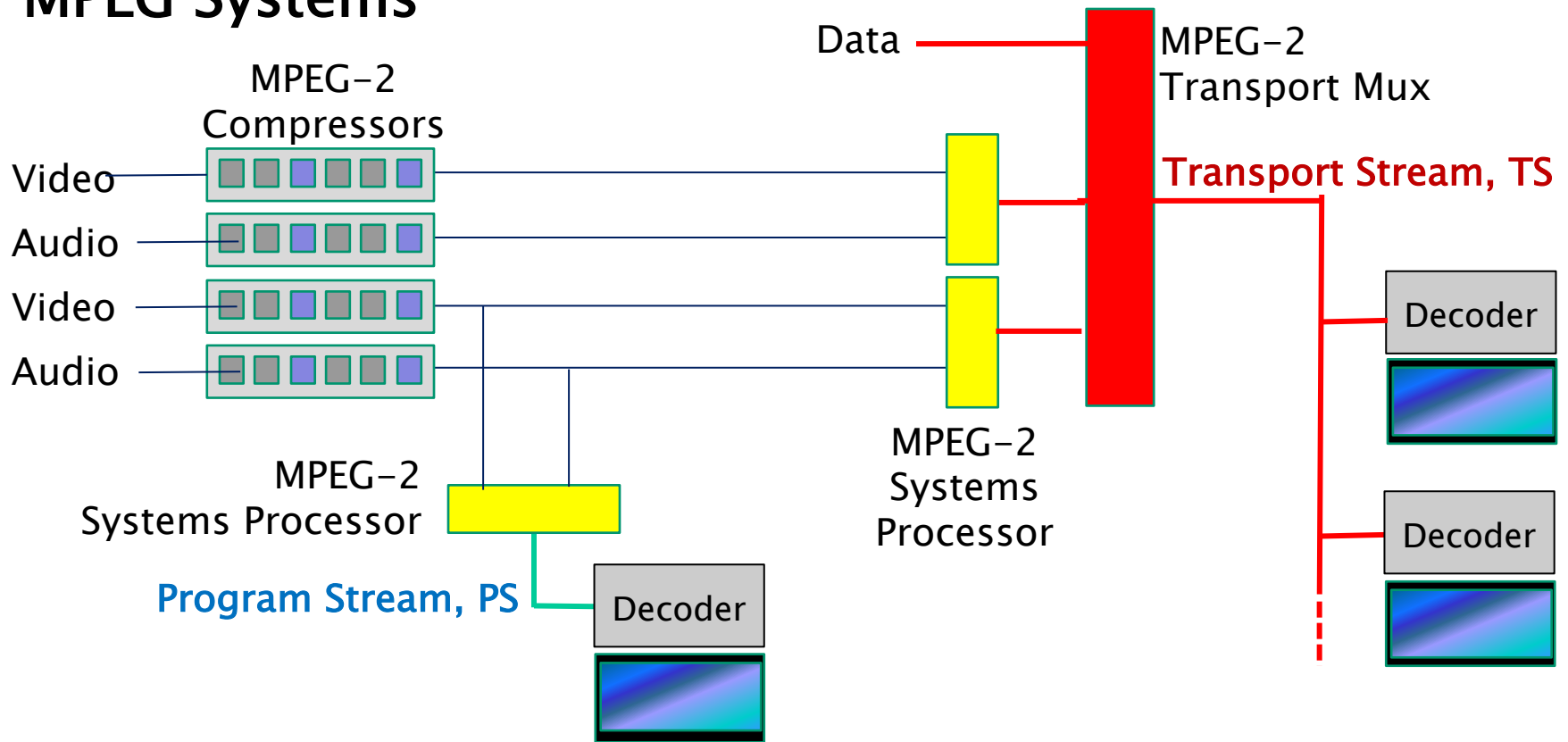
Web-based streaming (DASH, etc)

- ISO BMFF (as in mp4)

TS and PS 

(ISO BMFF – see later)

MPEG Systems



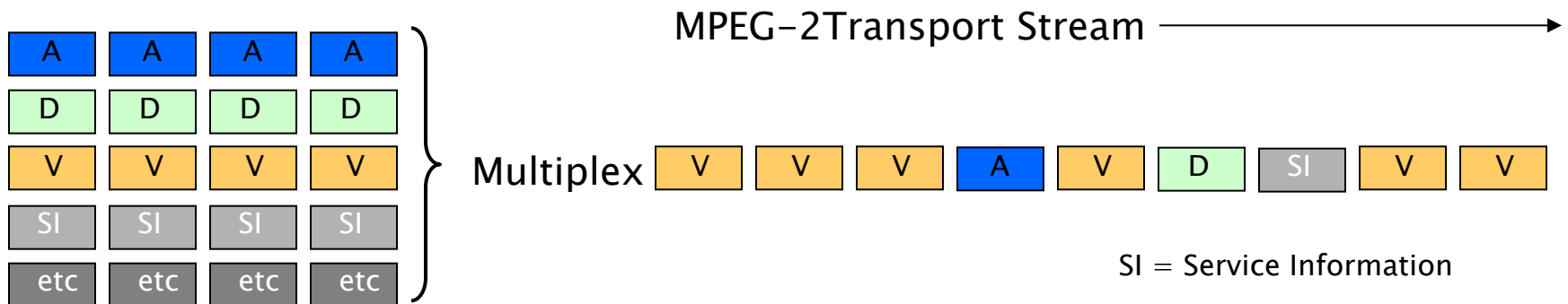
Requirement:

- To transmit Audio and Video compressed data in:
 - (i) reliable channels – studio, DVD, reliable media (PS)
 - (ii) transmission in hostile environments (TS)
- Allow interchange between these two formats

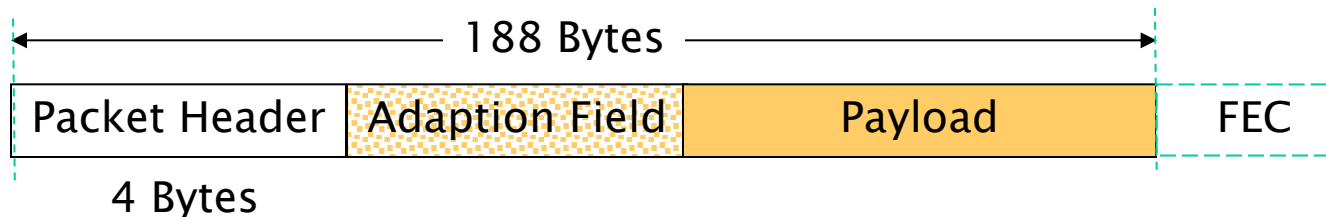
Transport Stream Multiplex

Transport Stream: Composed of A number of Program Streams (and their Elementary Streams)

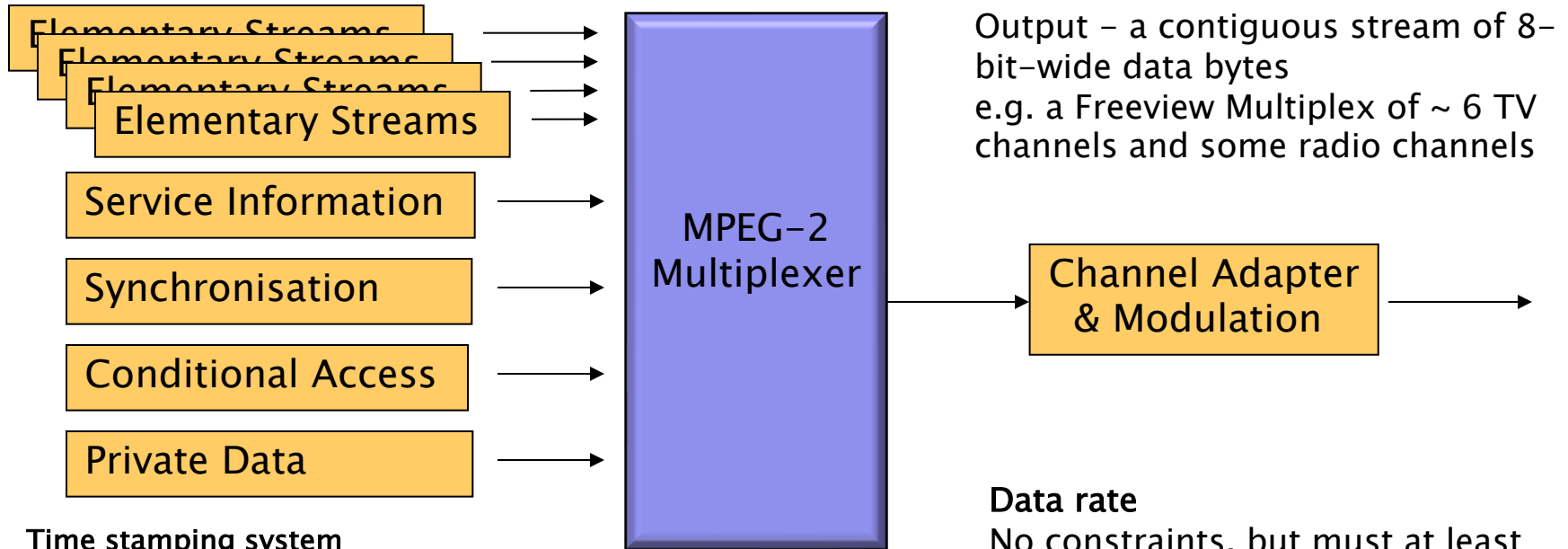
- when MPEG-2 content is to be distributed / networked, Transport streams are the structure
- they are a multiplex of PESs
- i.e. carry a number of programmes



Transport Stream Multiplex consists entirely of short, fixed length transport packets



Transport Stream Multiplex



Time stamping system

to ensure that related elementary streams are replayed in synchronism at a decoder

Tables of service information

network parameters, programmes within the multiplex and the nature of the various elementary streams

Conditional Access

Allow control of scrambling of one or more elementary streams (but CA not specified by MPEG)

Private Data

i.e. not spec'd by MPEG – include network management data, teletext, etc

Data rate

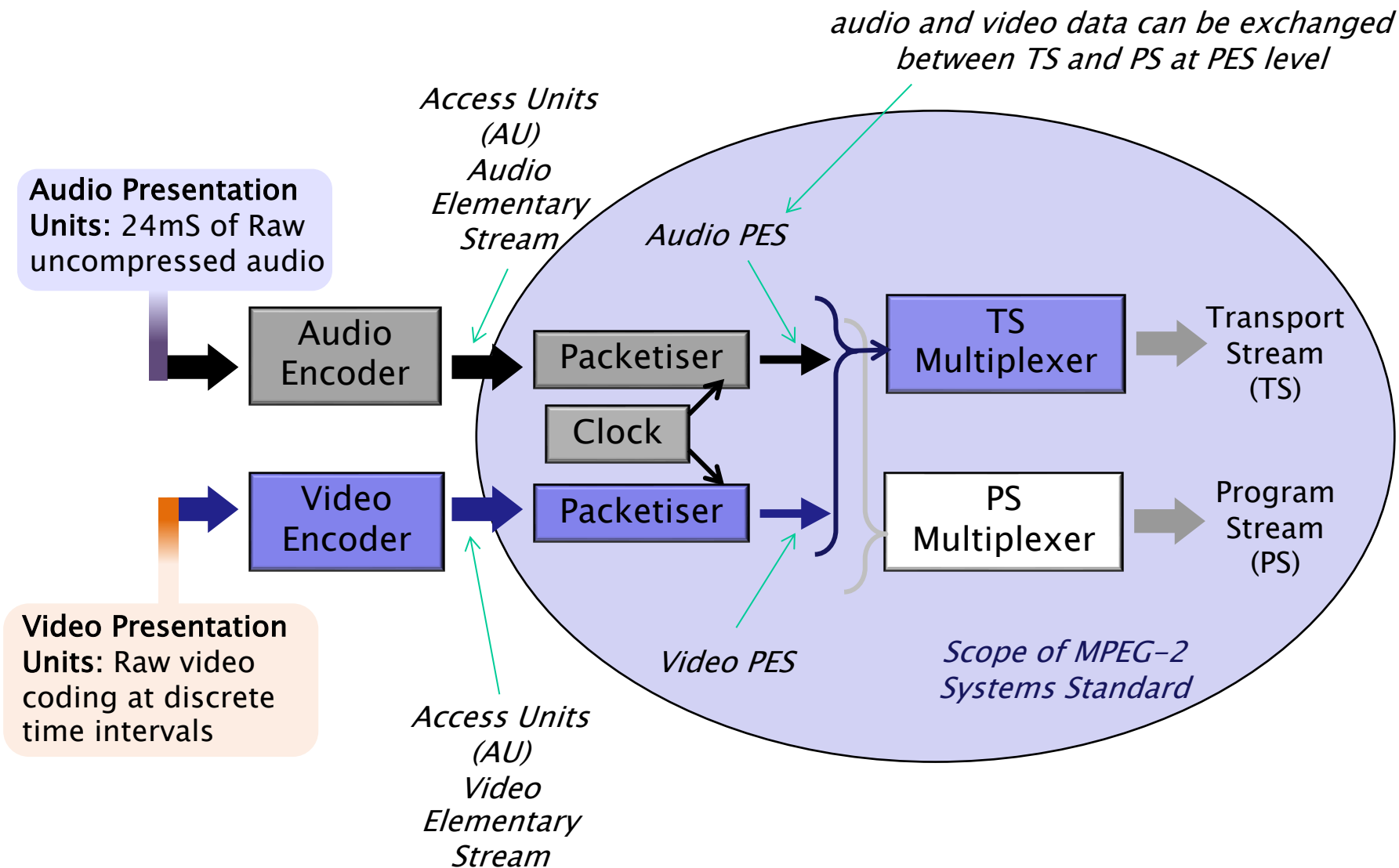
No constraints, but must at least equal the total of the combined data rates of the contributing elementary streams

Fixed or variable data rate; may contain fixed or variable data rate elementary streams

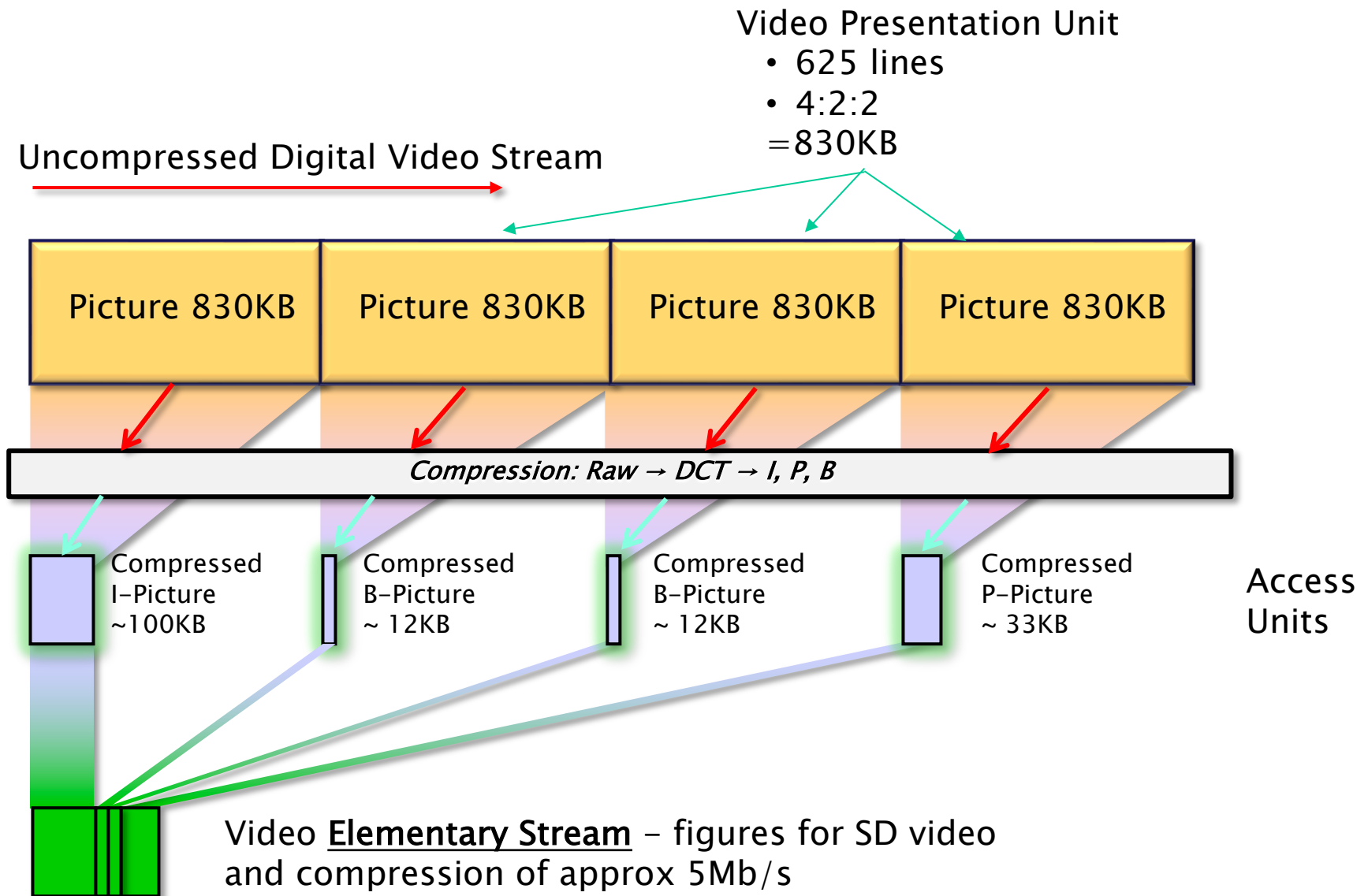
Not specified:

error protection, modulation technique, physical layer technologies

MPEG-2 Packets and Multiplexes

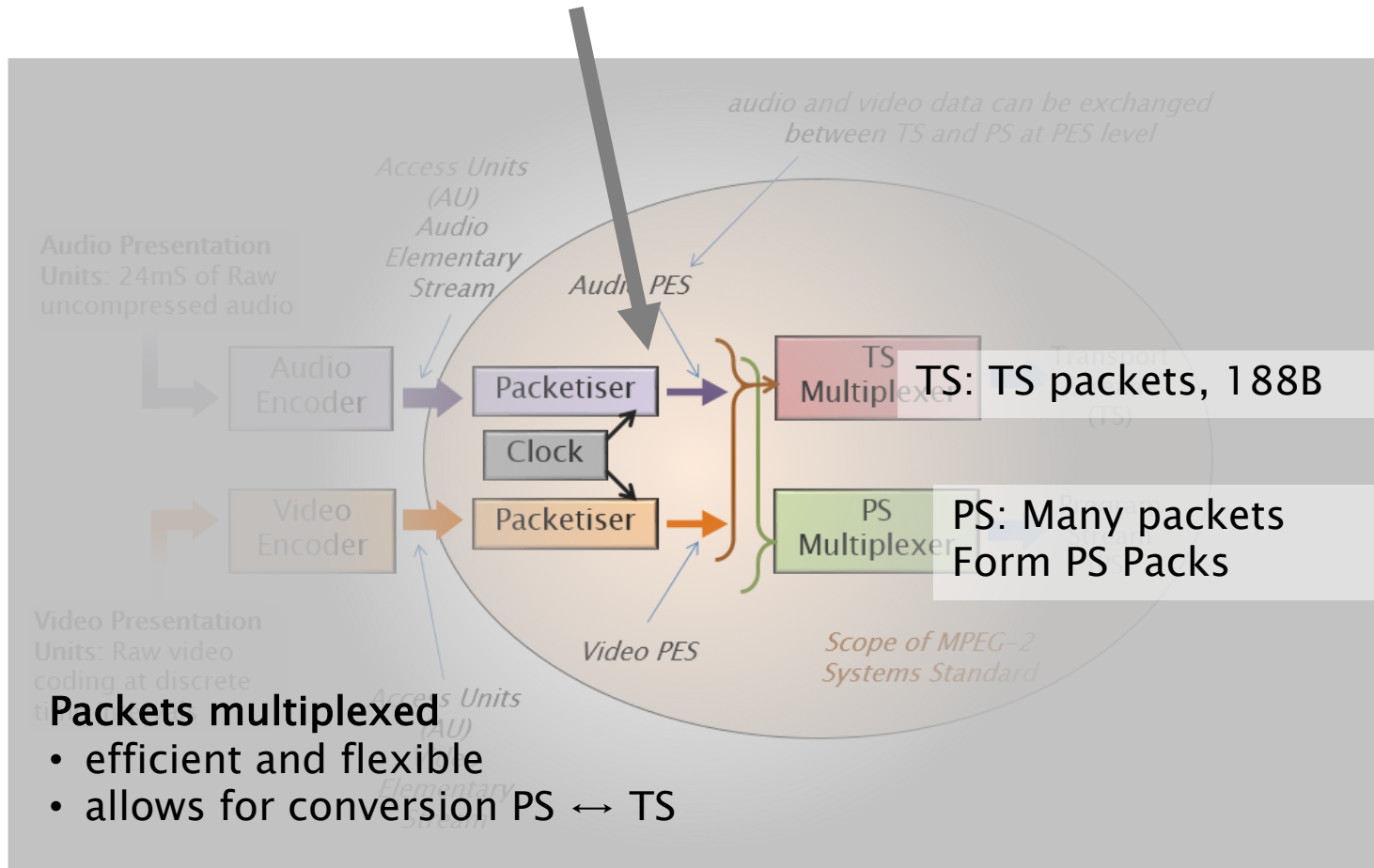


Video Elementary Stream

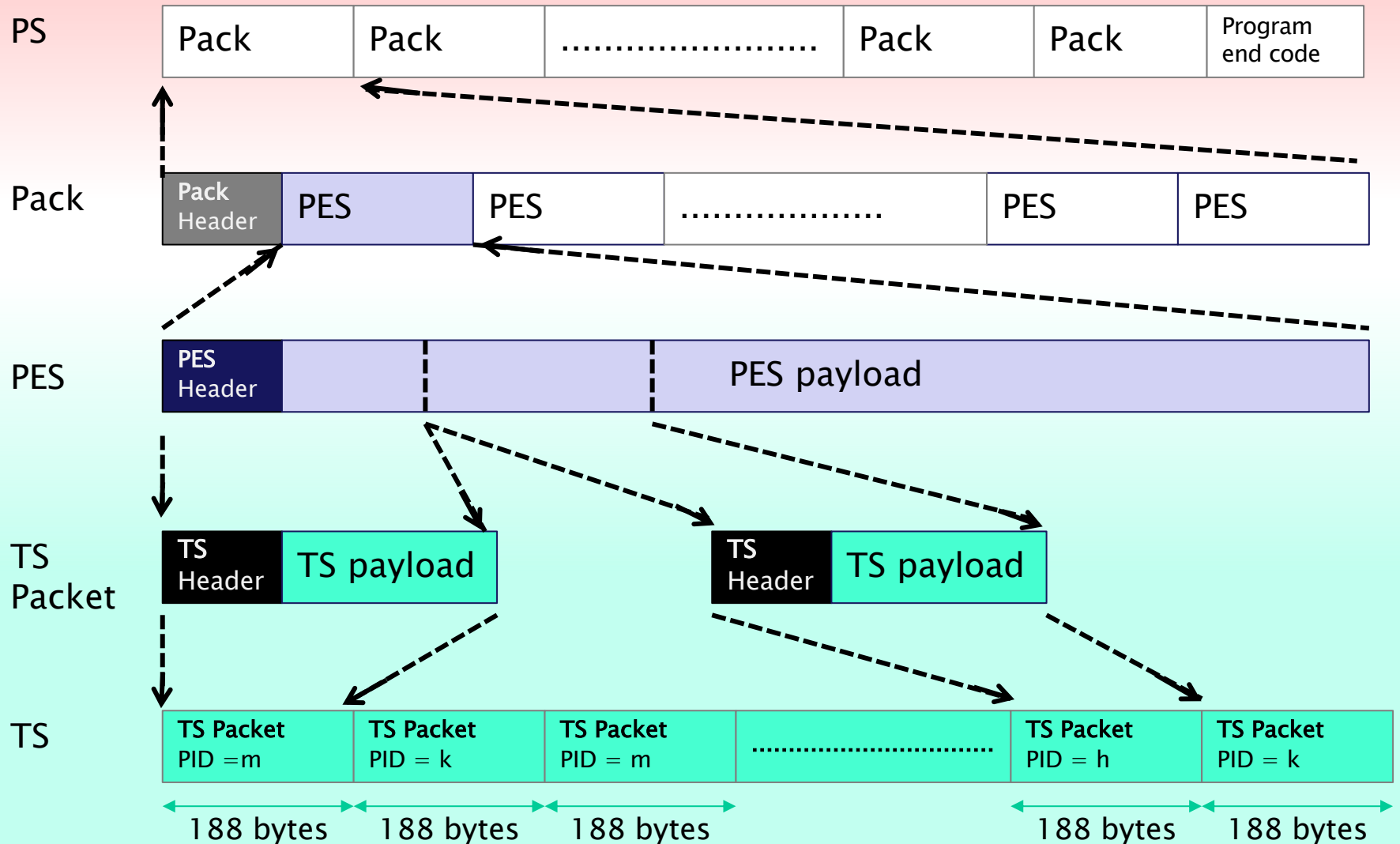


Packetizing

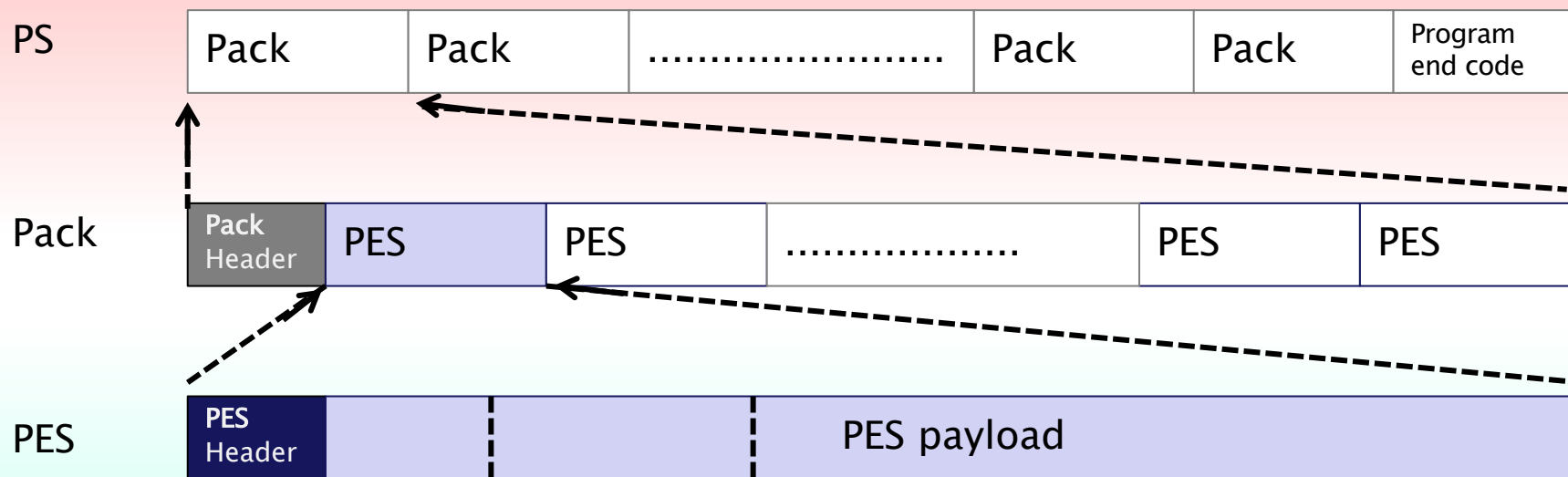
- Elementary Stream – ES – sub-divided – packetised
⇒ "Packetized Elementary Stream" = PES
- fixed-size chunks without regard for access unit boundaries



MPEG-2 – PES ⇒ PS Packs and TS Packets



MPEG-2 – PES ⇒ PS Packs and TS Packets



Program Stream – PS

- for transmission and storage in error-free channels
- **PES** – long variable-length packets
 - ... into variable length ***packs***
 - multiple PES packets
 - ... of nearly equal length (video and audio)

TS
Packet

TS

MPEG-2 – PES ⇒ PS Packs and TS Packets

PS

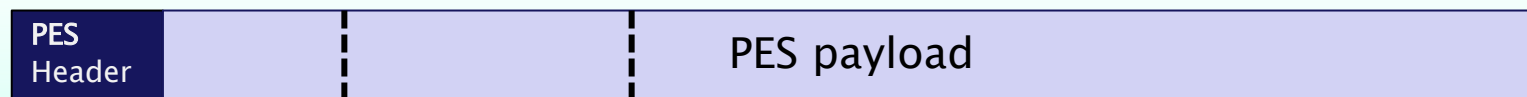
Transport Stream – TS

- for channels with likely errors
- short fixed-length: 188B

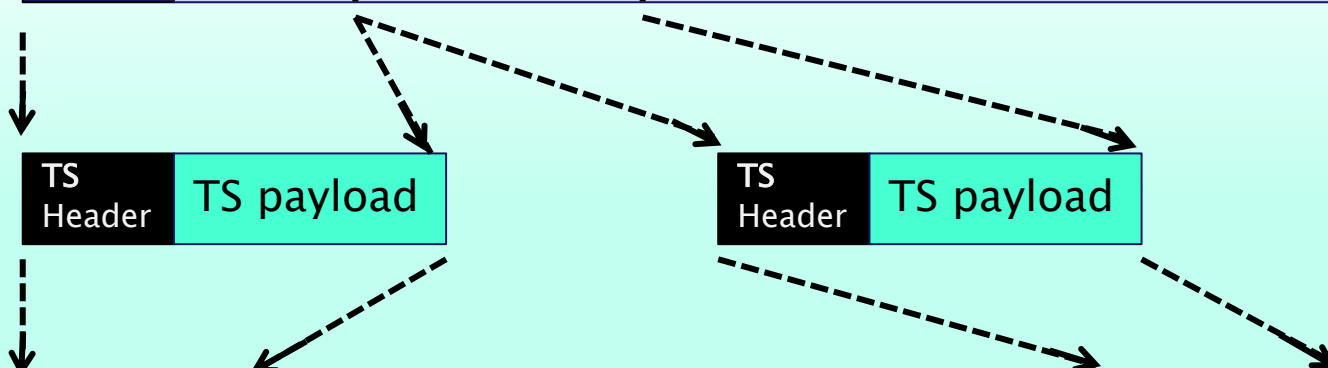
Pack

- PES packets – long variable length
 - Sub-divided
 - carried as payload of TS packets

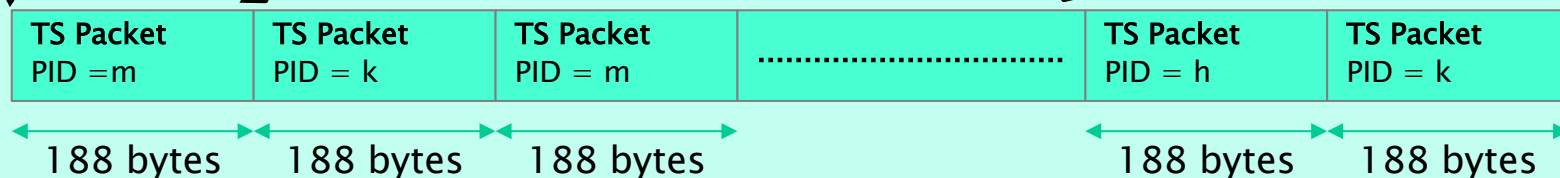
PES

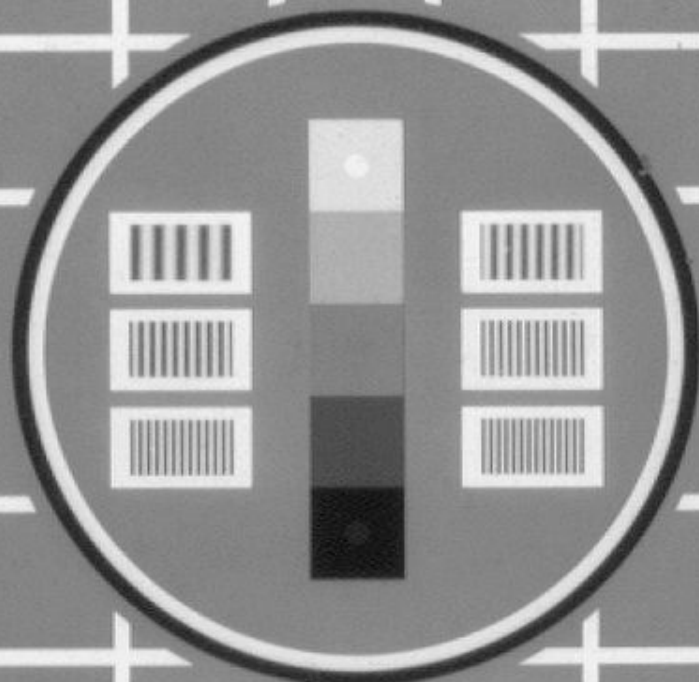


TS
Packet



TS





• | D | •

BBC 1 405 LINES

DVB

DVB – Digital Video Broadcasting



Industry Organisation

Standardisation Process –

Joint Technical Committee (JTC) of:

- ETSI – European Telecommunications Standards Institute
 - CENELEC – European Committee for Electrotechnical Standardization CENELEC
 - EBU – European Broadcasting Union
- Note standards refer to other standards and patents, e.g. MPEG

Standards

Satellite	DVB-S, DVB-S2, SH
Cable	DVB-C, DVB-C2
DTT	DVB-T, DVB-T2
DTT for handhelds	DVB-H, DVB-H2
IPTV	DVB-IPTV
DASH	DVB-DASH

DVB-3DTV – Frame Compatible Plano-Stereoscopic 3DTV
DVB-C2 – 2nd Generation Cable Transmission
DVB-CID – Carrier Identification for Satellite Transmission
DVB-CPCM – Content Protection and Copy Management
DVB-FF – DVB File Format
DVB-GEM – Globally Executable MHP
DVB-GSE – Generic Stream Encapsulation
DVB-H – Broadcasting to Handheld Devices
DVB-IPDC – Internet Protocol Datacasting
DVB-IPTV – Internet Protocol Television
DVB-MHP – Open Middleware for Interactive TV
DVB-PCF – Portable Content Format
DVB-RCS2 – Return Channel Satellite
DVB-RCT – Return Channel Terrestrial (03/2001)
DVB-S2 – 2nd Generation Satellite Broadcasting
DVB-SH – Satellite Services to Handheld Devices
DVB-T – Digital Terrestrial Television
DVB-T2 – 2nd Generation Terrestrial Broadcasting
DVB – Emergency Warning System
DVB Project – An Introduction
Hierarchical Modulation in DVB-T (03/2000)
plus many others (inc addenda, etc) ...

Digital Terrestrial TV (DTT)

Digital Video Broadcast, Terrestrial Television – DVB–T

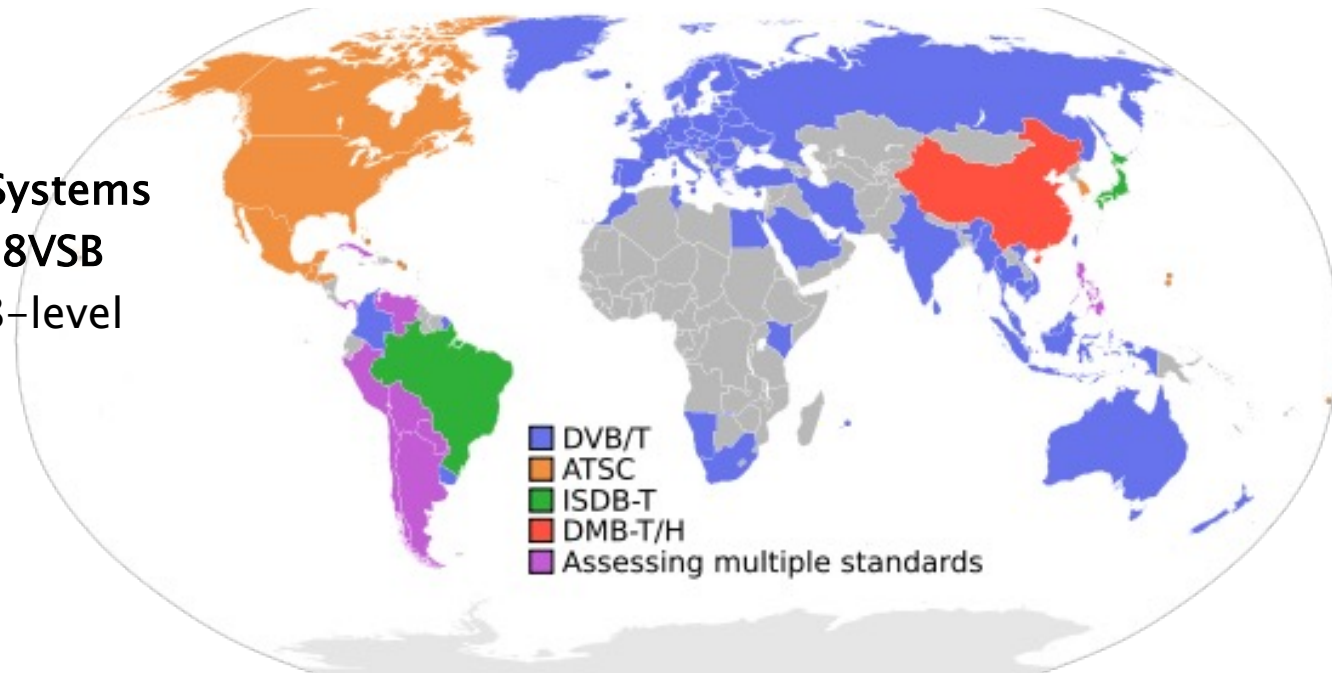
- Europe; also DVB–S for Satellite, DVB–C for Cable; DVB–H for handhelds
- MPEG2 format; also

Integrated Services Digital Broadcasting, Television – ISDB–T

- Japan; also ISDB–S for Satellite, ISDB–C for CableTV
- MPEG2 format
- Modulation: PSK, COFDM/PSK/QAM

Advanced Television Systems Committee – ATSC–8VSB

- Modulation: 8VSB: 8–level Vestigial Sideband
- American Continent
- MPEG2–TS



DVB Specifies ...

Specifies – essentially a layer one protocol for transport with Data Link (Transport Streams, GSE, etc)

Including –

Input Interface

- usually Transport Streams – i.e. TV
- Generic Stream Encapsulation – e.g. IP

Modes

- Variable Coding and Modulation
- Adaptive Coding and Modulation

FEC

- e.g. DVB-T2: LDPC, BCH $\frac{1}{2}$, $\frac{3}{5}$, $\frac{2}{3}$, $\frac{3}{4}$, $\frac{4}{5}$, $\frac{5}{6}$

Modulation

- e.g. DVB-T2: OFDM

Modulation Schemes

- Q/8/16/32PSK, QAM
- e.g. DVB-C2: 16–4096QAM

Guard Interval

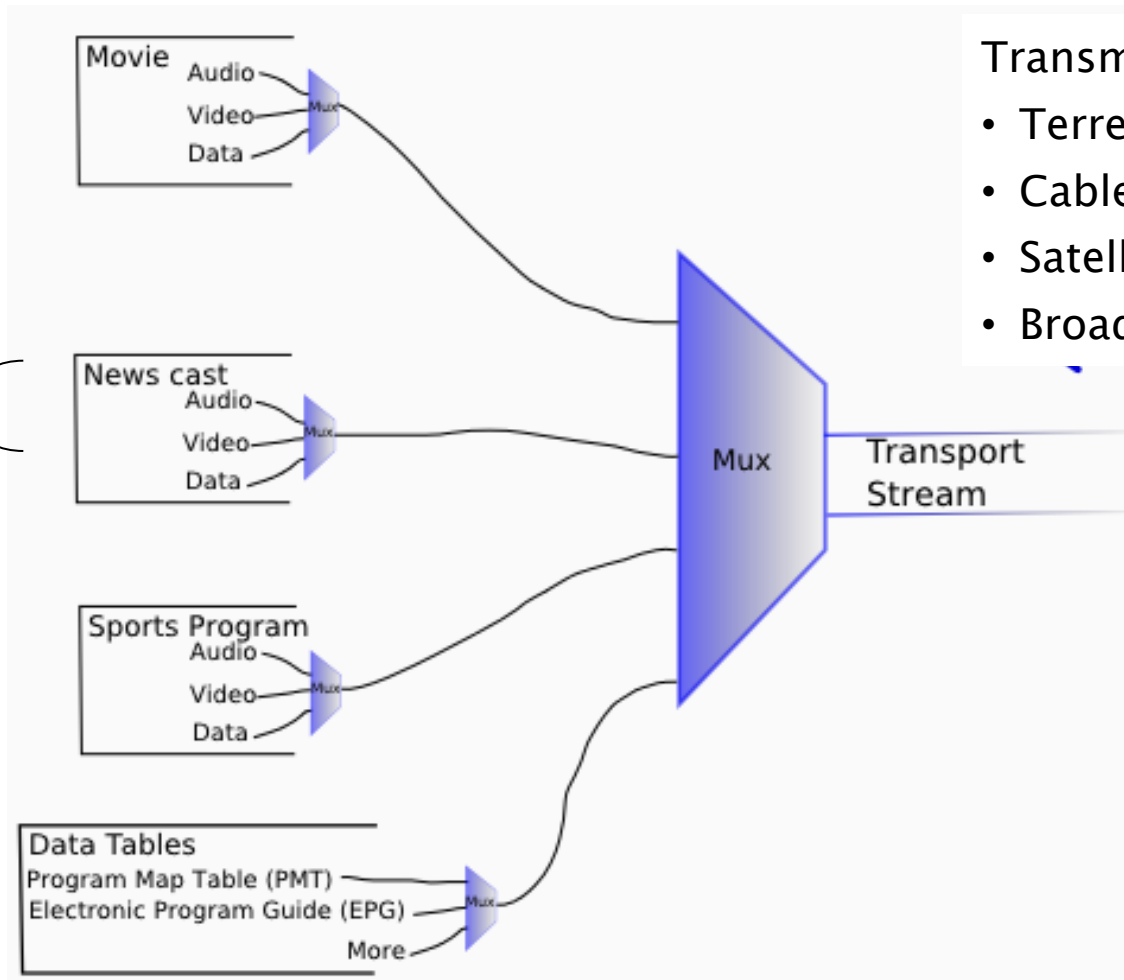
- e.g. DVB-T2: $\frac{1}{4}$, $\frac{19}{256}$, $\frac{1}{8}$, $\frac{19}{128}$, $\frac{1}{16}$, $\frac{1}{32}$, $\frac{1}{128}$ (fractions of a symbol)

Interleaving

- Bit– Time– and Frequency–Interleaving

Transport Stream Multiplex

Program
Streams



Transmission:

- Terrestrial
- Cable
- Satellite
- Broadband

Compression

Why Compress Video

- Size of uncompressed 1080p HDTV
 - $1920 * 1080 \approx 2 \text{ Mpixels per frame}$
 - For RGB this is 6MB per frame
 - Thus 1 second of video needs
 - $25 * 6\text{MB} = 150\text{MB}$
 - Data rate of $150 * 8 = 1200 \text{ Mbps} = 1.2 \text{ Gbps}$
 - One hour of video needs $3600 * 150 \text{ MB}$
- $= 540,000 \text{ MB} = 540 \text{ GB}$

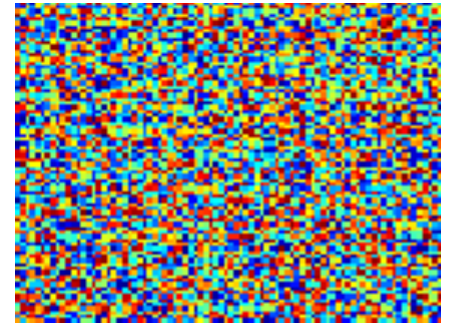
Why Compress Audio?

- CD quality = $2 * 44.1 * 16 \text{ Kbps} \approx 1.4 \text{ Mbps}$
 - About 1 / 850 of HDTV video
- 1 hour of audio needs $3600 * 1.4 / 8 = 630 \text{ MB}$
 - 100 tracks take about 360 minutes $\approx 4 \text{ GB}$

Compression – Information Theory

IF ...

- image is all the same colour
 - very easy to compress – just replace with colour value and codeword to indicate
- every pixel is a random number between 0 & 255 and not correlated in any way with other pixels
 - it is impossible to compress

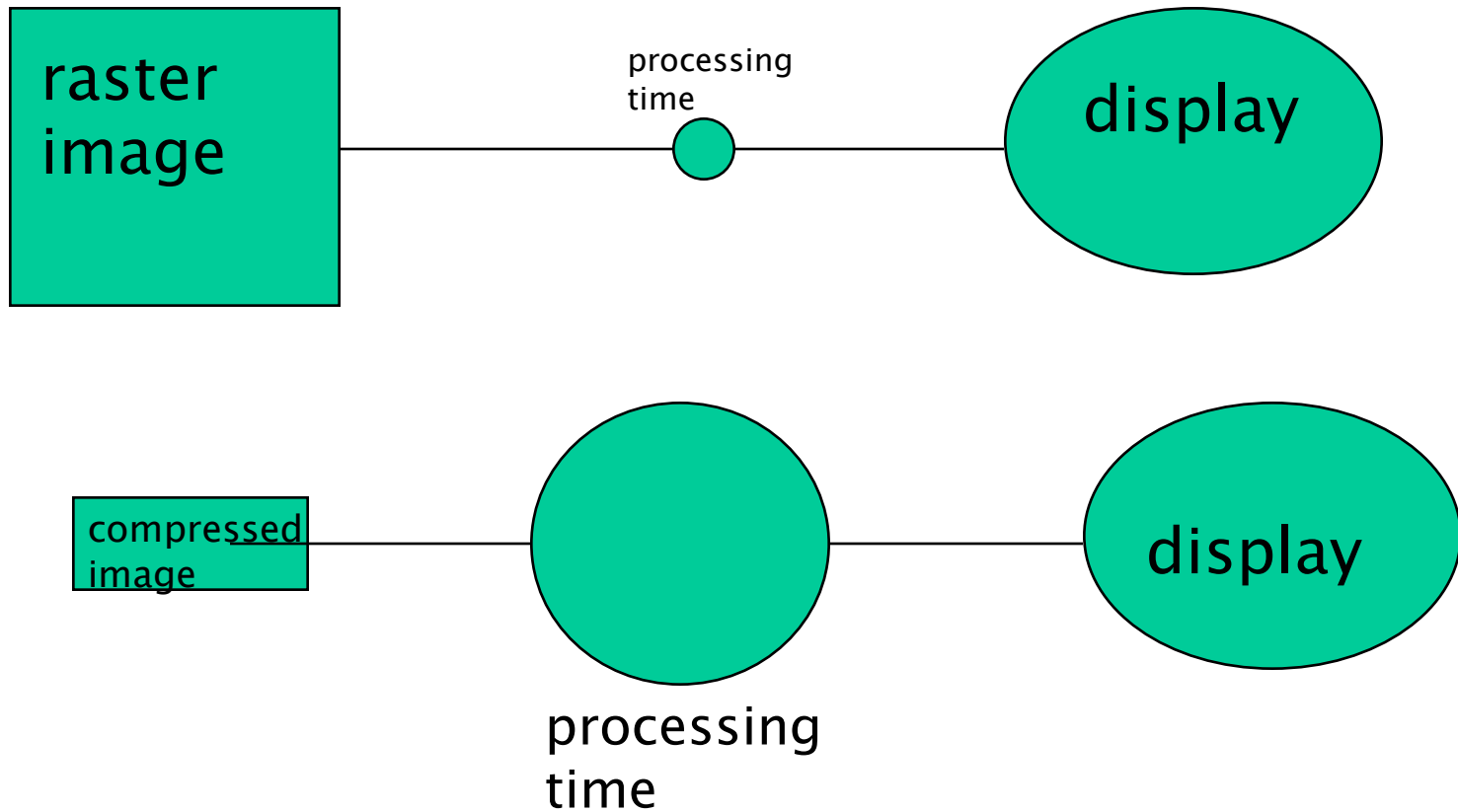


In between –

- look for patterns only for lossless compression
- additionally psychovisual characteristics for lossy compression



Compressed and Uncompressed Images



Lossless Compression (of Images)

- Three standard techniques that take advantage of regular and repeating data
 - Run Length Encoding (RLE)
 - Huffman Compression
 - Differential (or Predictive) Coding
- ironically in their pure form they are most suited to compression of bitmap representations of computer generated images

Lossless Compression

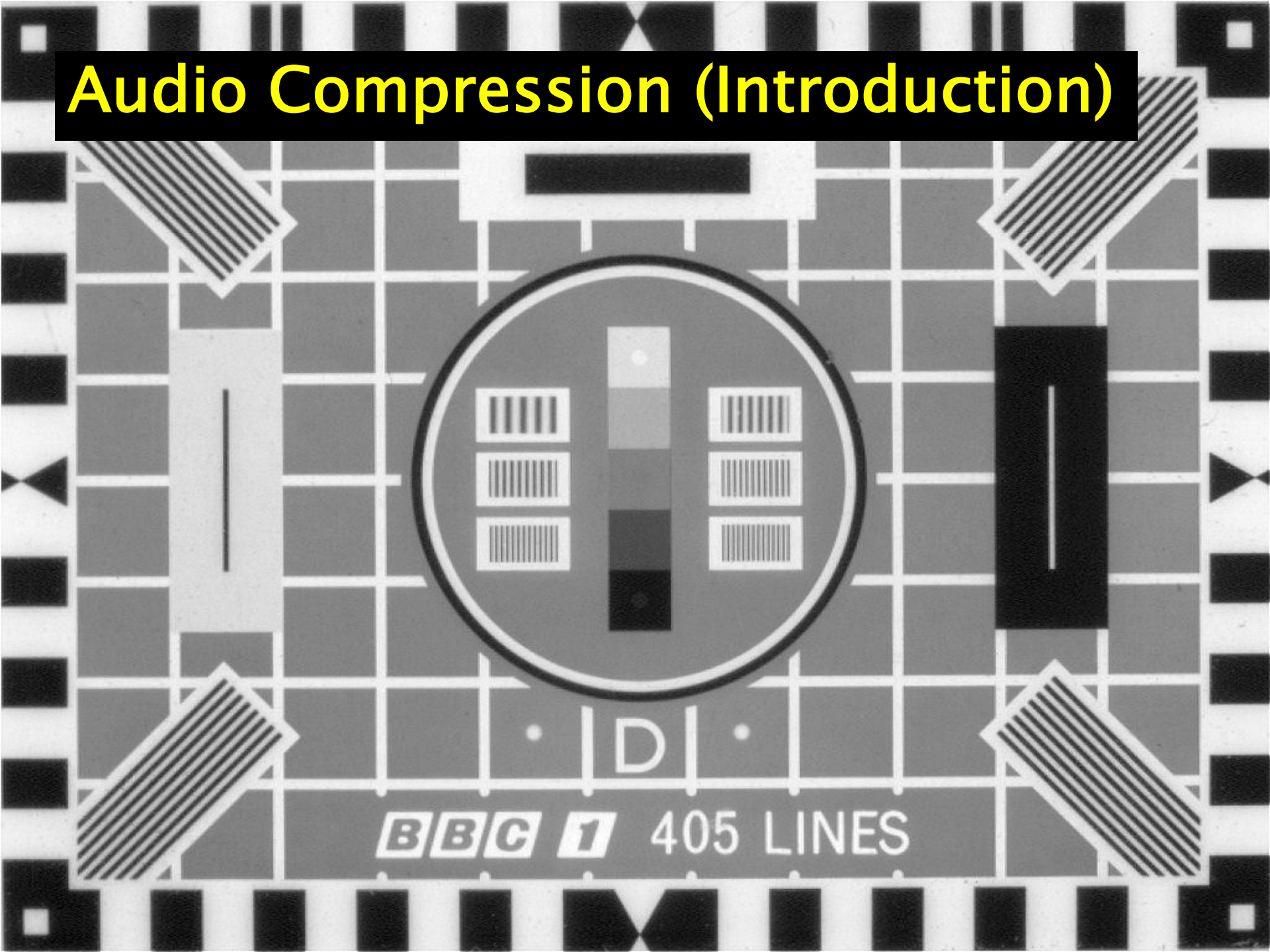
- RLE replaces repeating bytes with number and kind:
 - e.g. 777555555 becomes 3765
 - compression ratio $9:4 = 2.25$
- In most applications RLE files are further compressed using Huffman which scans the whole file to discover the most frequently occurring bytes. These bytes are then translated to bit form, for example using only 1 bit for the most frequently occurring byte , 2 bits for the next most frequently occurring and so on

Bit Rate Reduction Principles

Main process of bit-rate reduction:

- Remove lower priority information from the signal at encoder
... according to what we *perceive* in the image
- **Coding** – including information about the bit-rate reduction
... codec at time of image capture
- **Decode** – to assemble and play the compressed file
... codec at time of viewing image

Audio Compression (Introduction)



Audio Compression

Audio Compression

- Compression Usability, Latency
- Encoding Architecture
- Perceptual Coding: Psychoacoustic Models
- Transform Domain Methods, Decoding
- Coding, Bit Allocation, Frames

AAC – Advanced Audio Coding

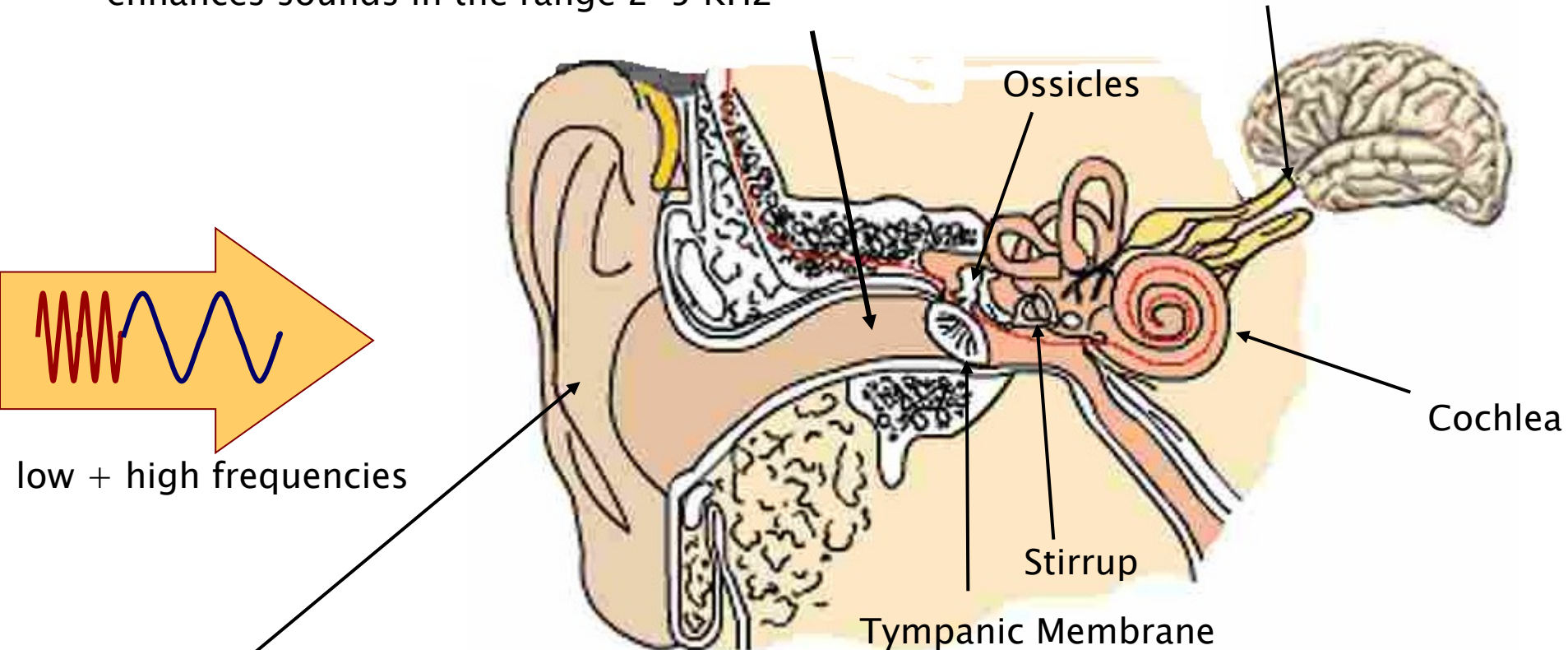
- MPEG-4 AAC, Low Delay Coder, Error Resilience,

Variable Bit Rate

The Physiology of Hearing

Auditory Canal

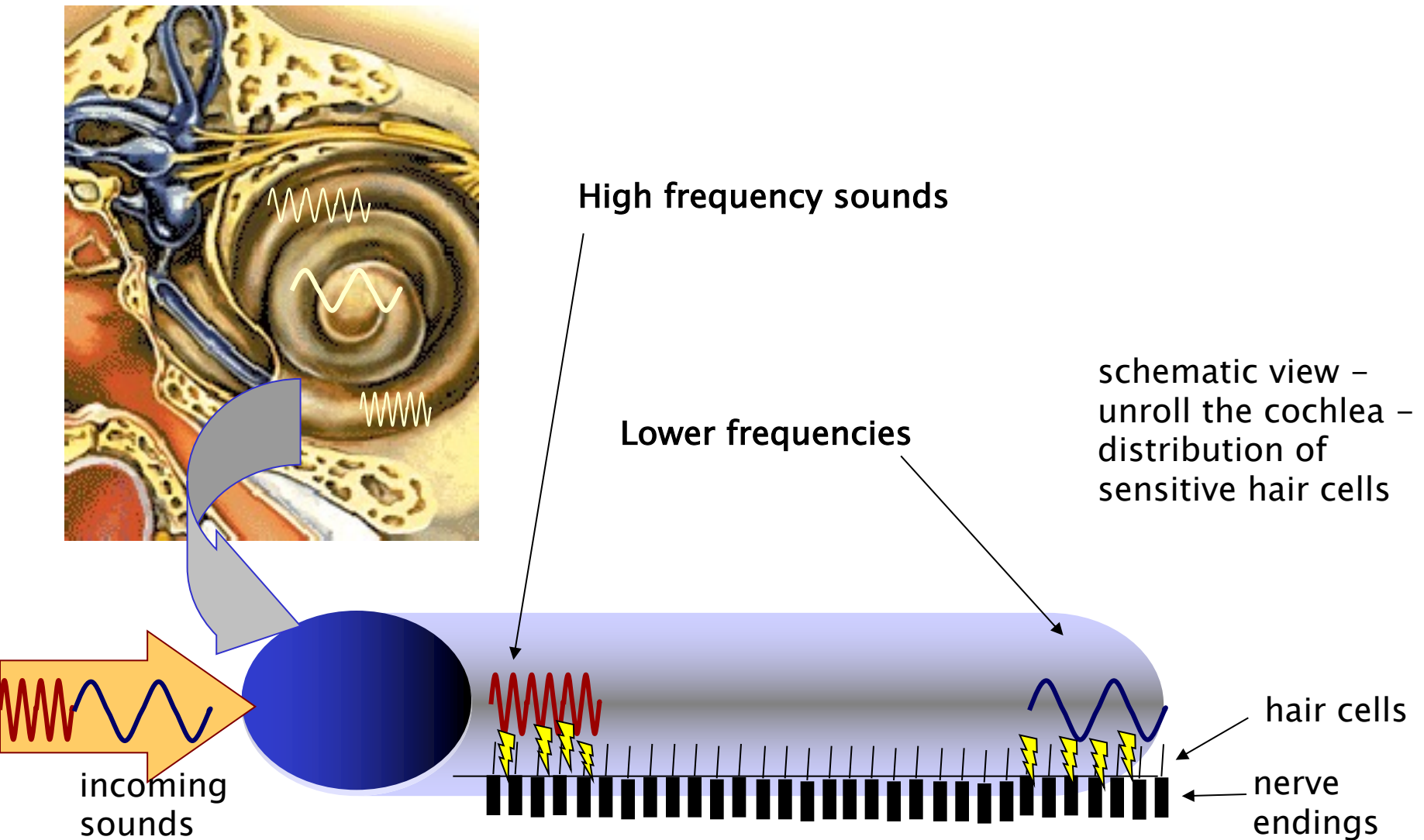
- acts as a closed tube resonator
- enhances sounds in the range 2–5 KHz



Pinna

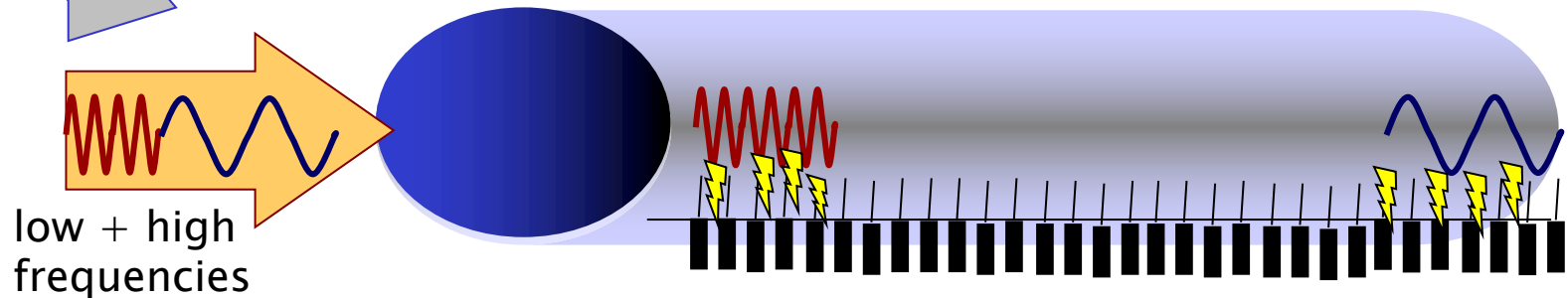
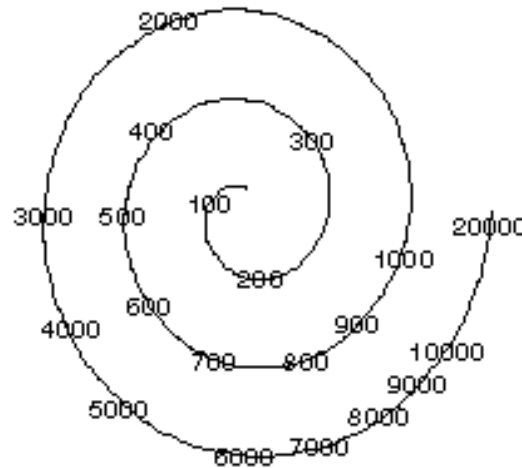
- presents larger area to received sound energy
- “collects” sound – like a preamplifier

The Physiology of Hearing – The Cochlea

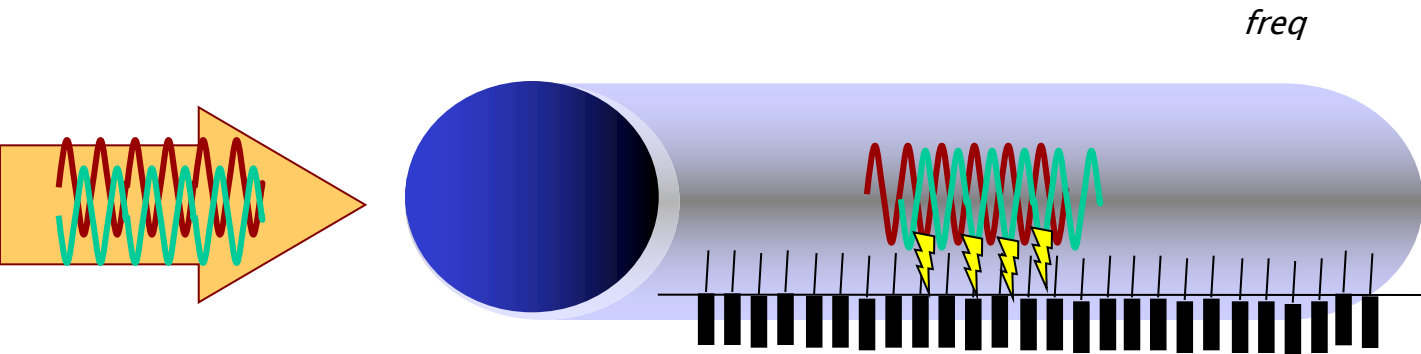


The Physiology of Hearing – The Cochlea

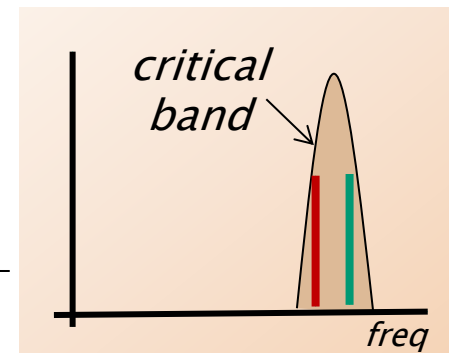
<http://www.hitl.washington.edu/publications/hollander/2.html>



The Physiology of Hearing

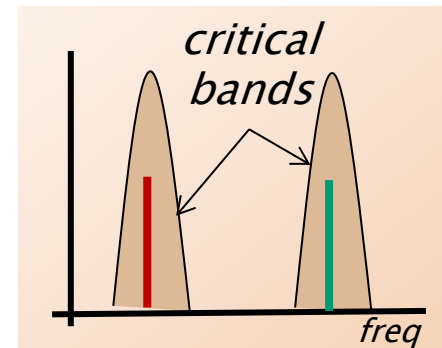
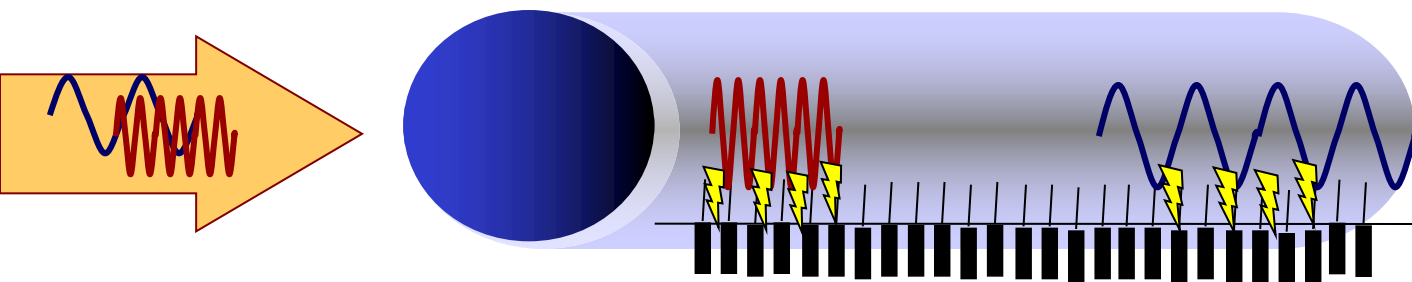


freq



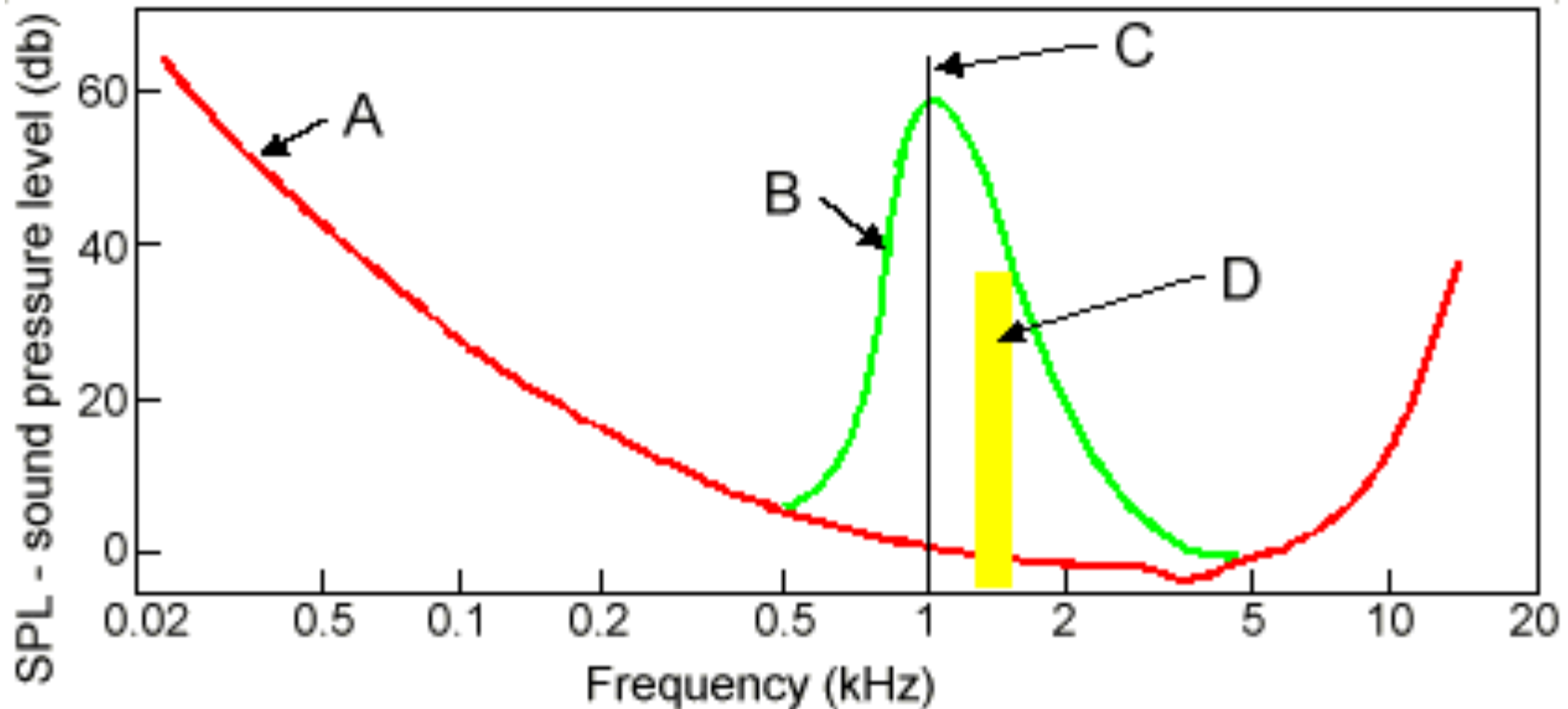
- Two signals within a critical band – closely related in frequency
- stimulate the same area of the membrane in the cochlea
 - these two sounds added together don't give twice the loudness

critical band



- Two signals – frequencies separated
- stimulate the different areas of the membrane in the cochlea
 - not competing for the same nerve endings
 - give an appreciable increase of perceived intensity

Threshold of Hearing, Masking Threshold



- A Normal threshold of hearing
- B Modified threshold due to tone C
- D Band of noise rendered inaudible by the presence of tone C

MPEG Psychoacoustic Models

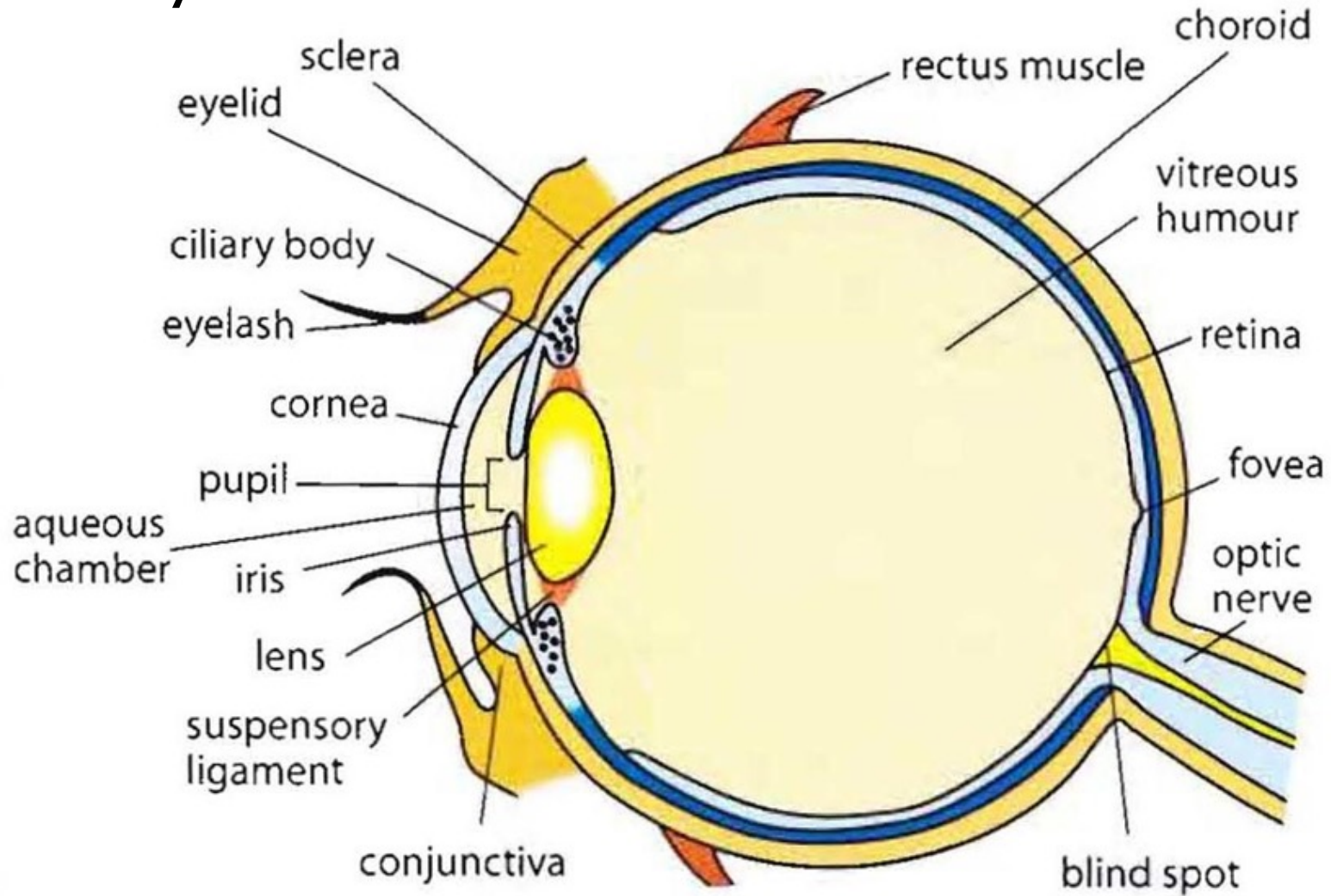
MPEG standard defines two models:

- **Psychoacoustic Model 1:**
 - simpler, less computationally expensive
 - serious compromises in what it assumes a listener cannot hear
- **Psychoacoustic Model 2:**
 - More features
 - more complex coding (assuming increased processor bandwidth)
 - Layer III (.mp3)

Video Compression (Introduction)



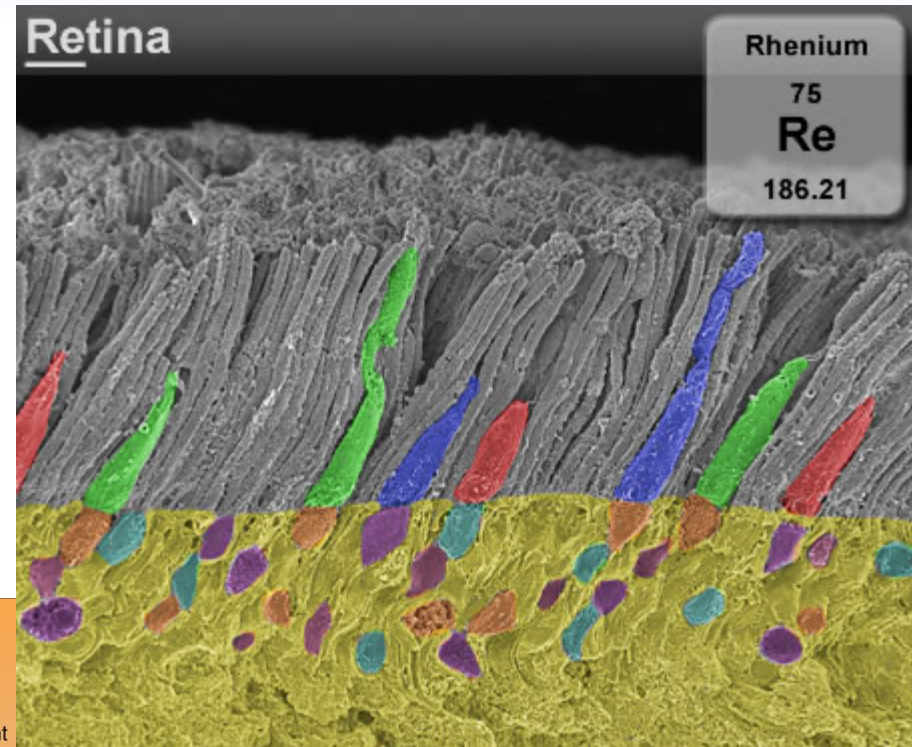
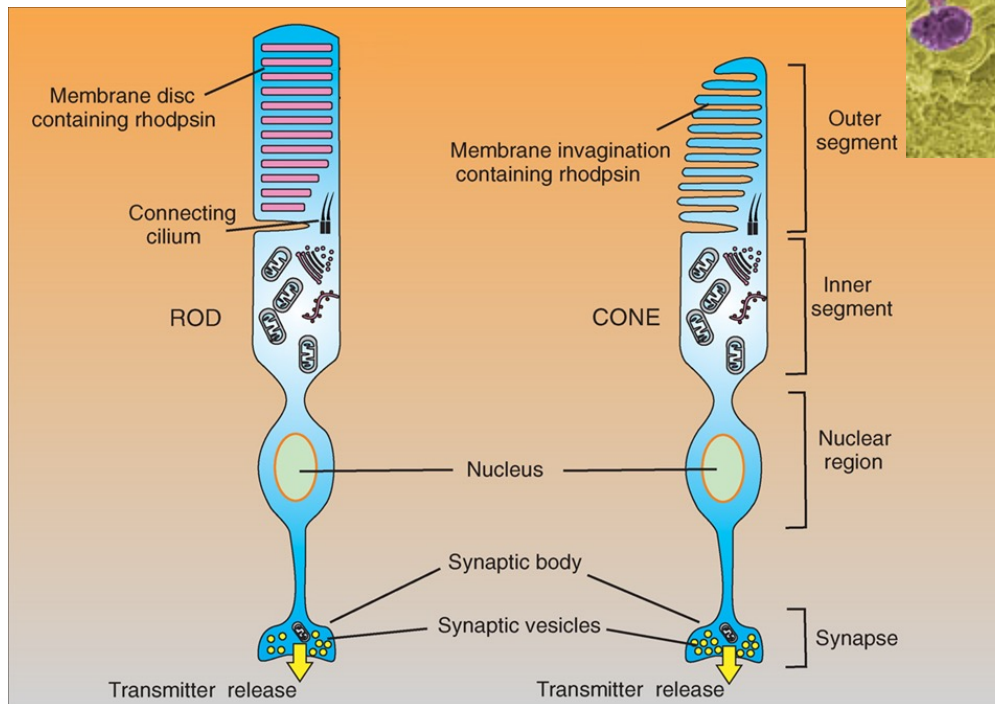
The Eye



Receptors

... Rods and Cones

http://www.cellsignallingbiology.org/csb/010/csb010fig10_rod_and_cone_structure.htm



Psychovisual – how we perceive images

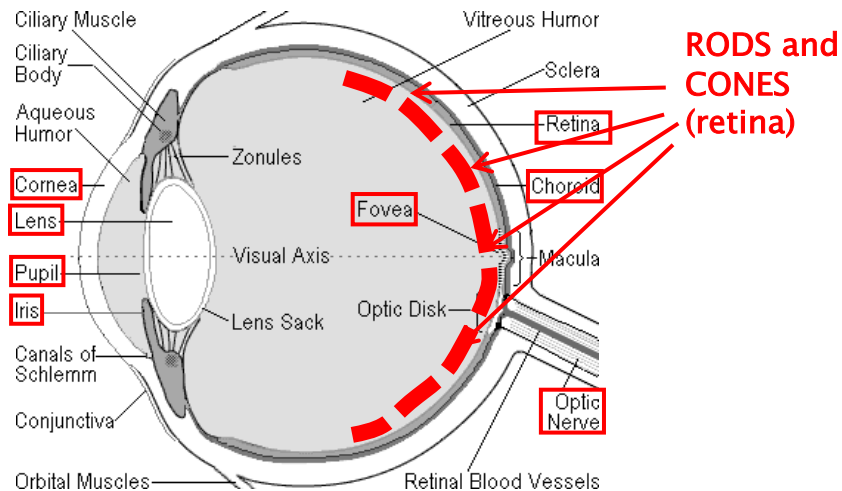
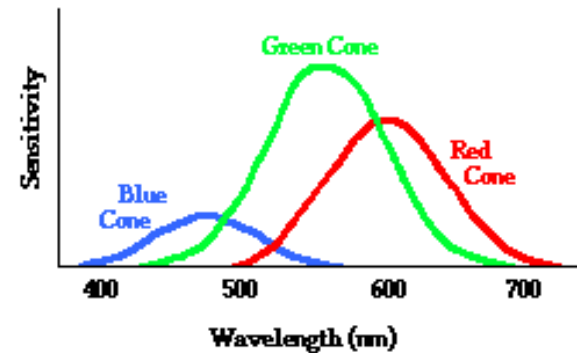
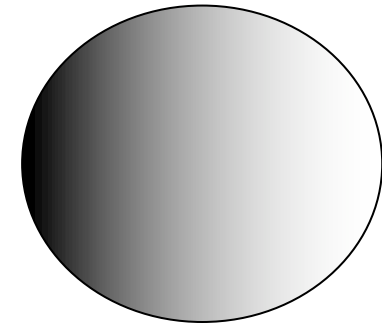
Light sensitivity in the eye

Rods – light intensity – not colour

- ~125 million across the retina
- characteristics of high speed black & white film

Cones – colour – responsive in three wavebands

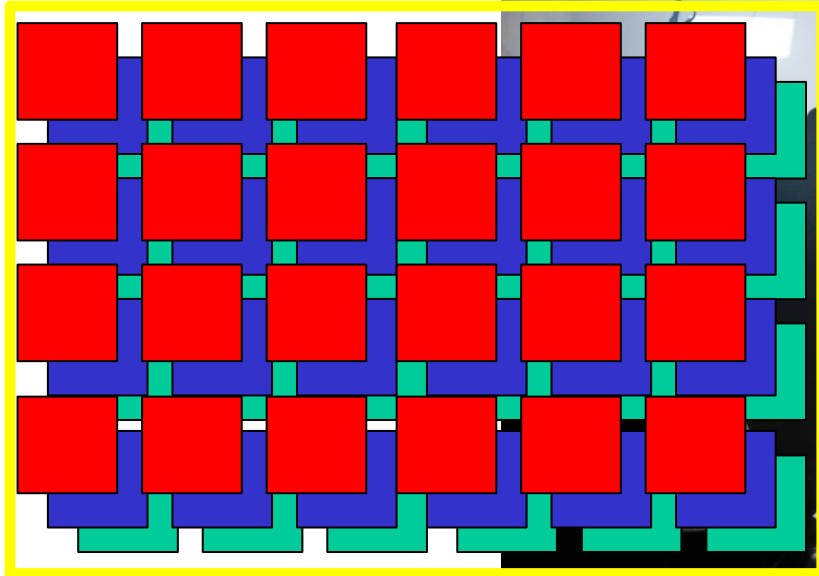
- 6 or 7 million cones
- characteristics of low-speed colour film



... Chroma Sub-sampling

Images Captured on a Camera

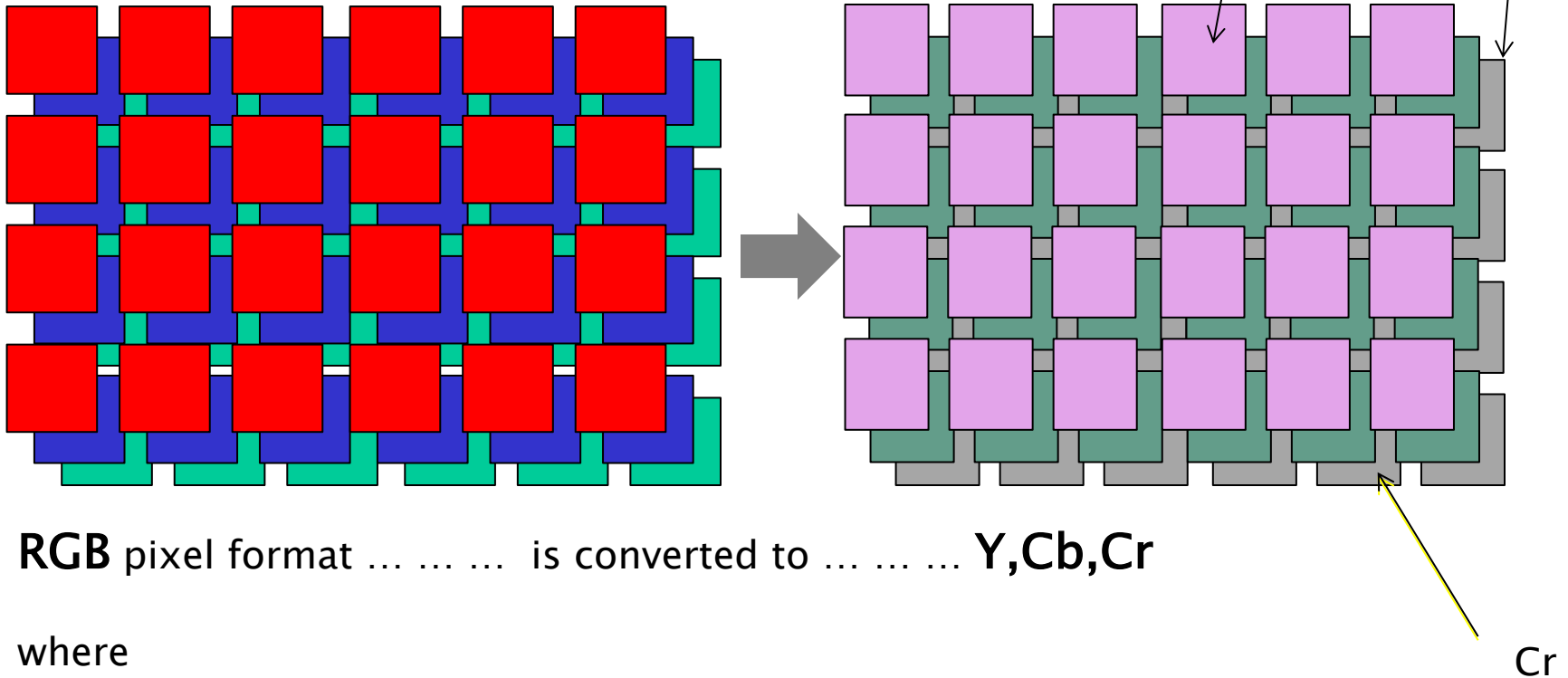
Original captured
image at camera
sensor:



- Image captured on red blue and green – responsive photo-sites
- each pixel is represented by R, B, G
- each pixel has its own value of light intensity

RGB to Y,Cb,Cr

During the *encoding* stage ...



RGB pixel format is converted to Y,Cb,Cr

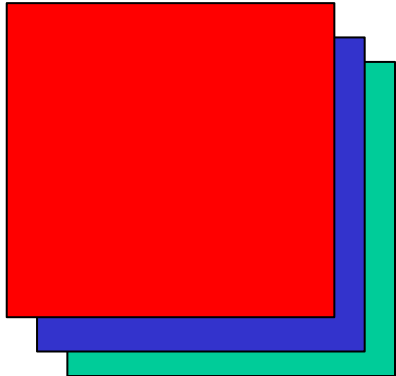
where

Y = luminance (light intensity)

Cb, Cr = colour information

Weighted RGB converted to Y Cb, Cr
 $0.2989 \cdot R + 0.5870 \cdot G + 0.1140 \cdot B$

RGB to Y,Cb,Cr – each pixel

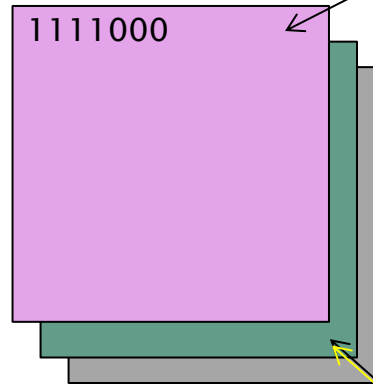
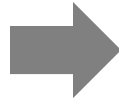
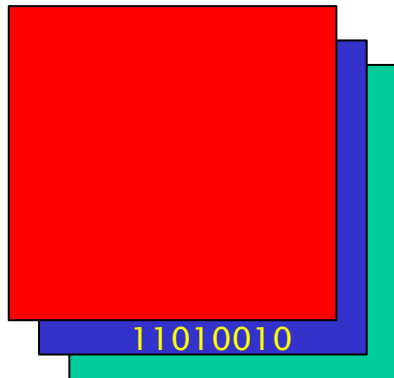


Overall luminance
i.e. greyscale, monochrome
of this pixel
= Y, ... i.e. the *luma*

Weighted RGB luminance:
approx 29% Red + 59% Green + 11% Blue



RGB to Y,Cb,Cr – each pixel



$$Cb = B - Y$$

- i.e. 8 bits of Cb (0–255)
- value of Cb:
value of original Blue
minus
value of luma

$$\text{Pink} = \text{Blue} - \text{Grey}$$

e.g.

$$120 = 210 - 90$$

01011010

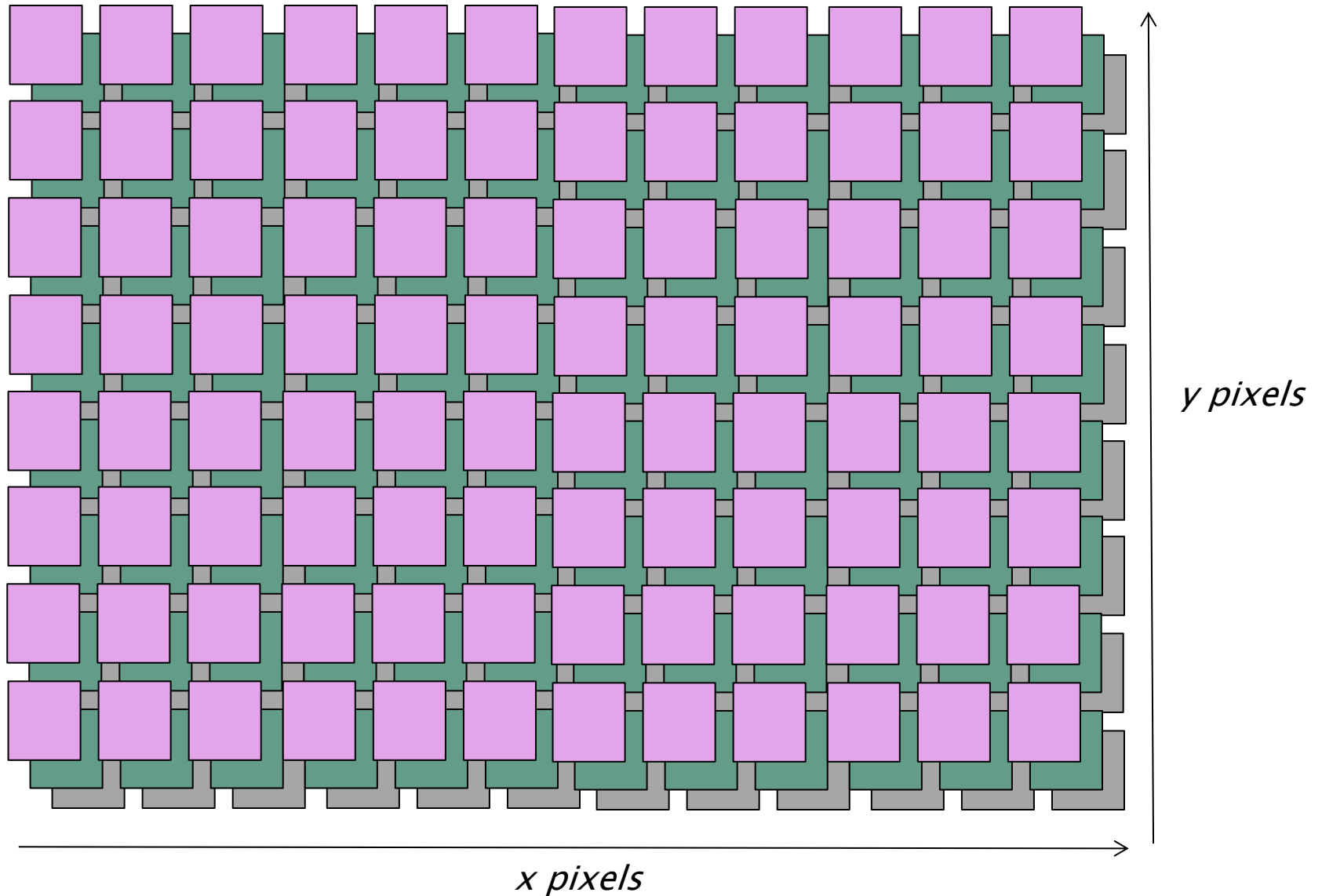
Overall luminance
i.e. greyscale, monochrome
of this pixel
= Y, ... i.e. the *luma*

Weighted RGB luminance:
approx 29% Red + 59% Green + 11% Blue

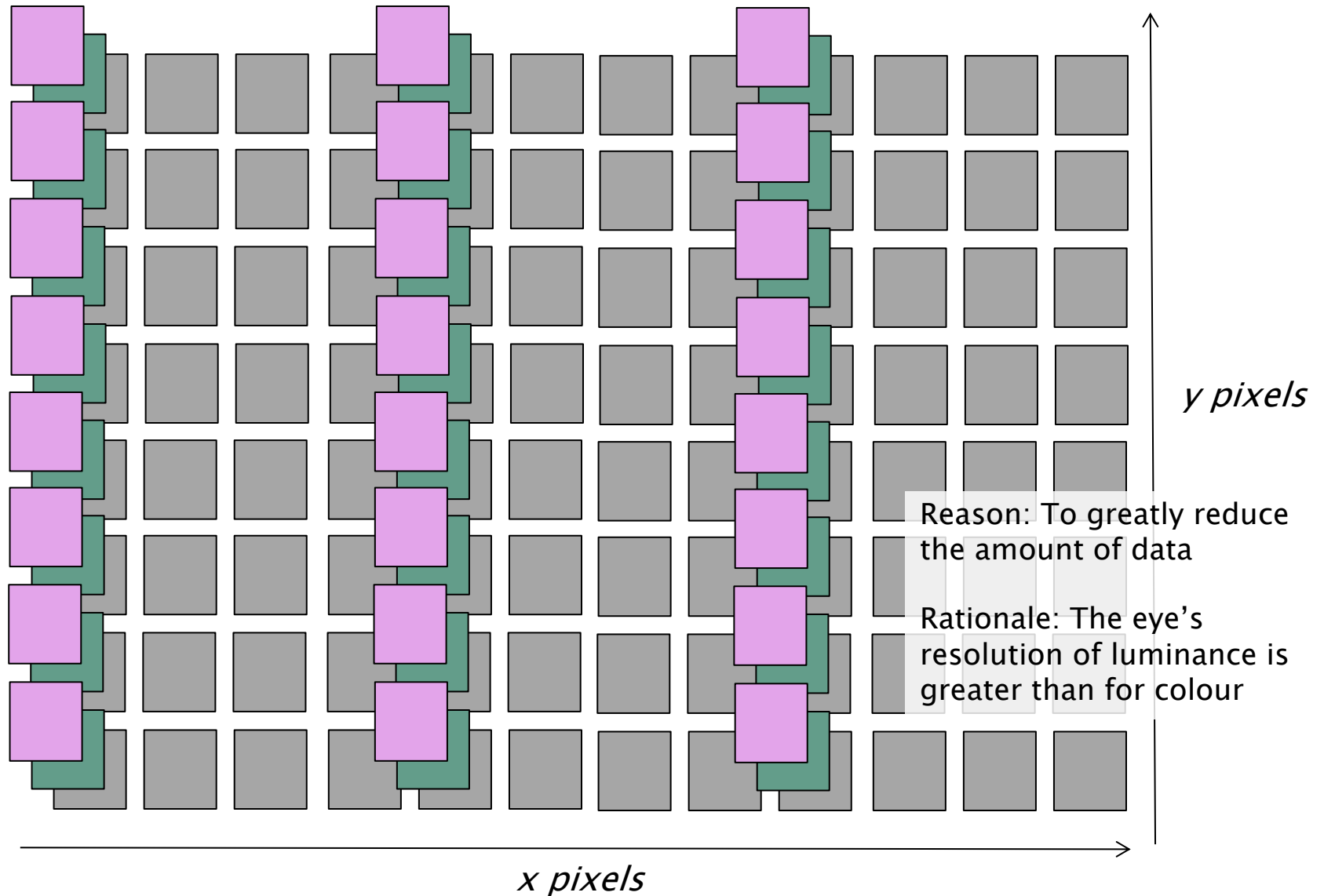
similarly,
$$Cr = R - Y$$

$$\text{Green} = \text{Red} - \text{Grey}$$

Y,Cb,Cr – for image of $x \times y$ pixels in “4:4:4”



Y,Cb,Cr – Chroma Sub-sampled to “4:1:1”

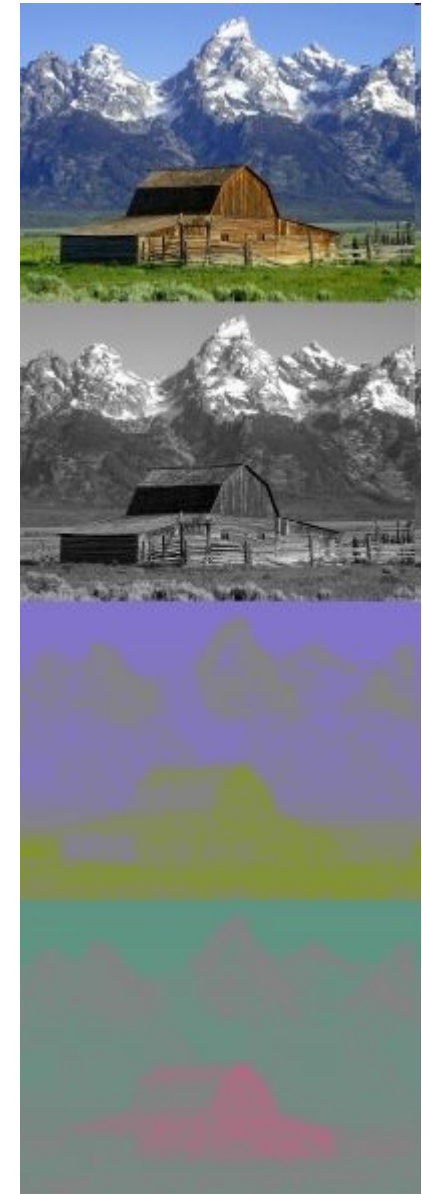


Y,Cb,Cr

Original image

YCbCr

- Colour Space: Y = Luminance Information; Cr, Cb = Chrominance Information
- Derived from RGB (gamma corrected)
- Y is dominant information; RGB colour info derived from Y, Cr, Cb relationships
 - (simplified): $Cb = B - Y$; $Cr = R - Y$



RGB to Y,Cr,Cb

i.e. there are now
three mappings of
the original RGB
image:

Y: Luminance
intensity

Cb: Intensity of the
values of
[B minus Y]

Cr: Intensity of the
values of
[R minus Y]



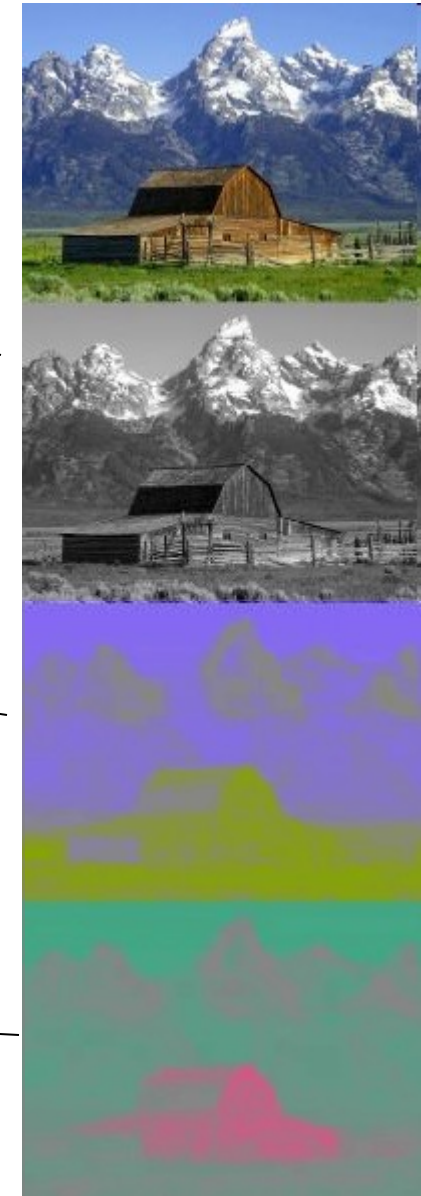
Y



Cb



Cr

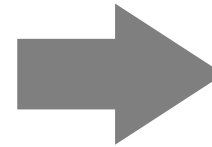
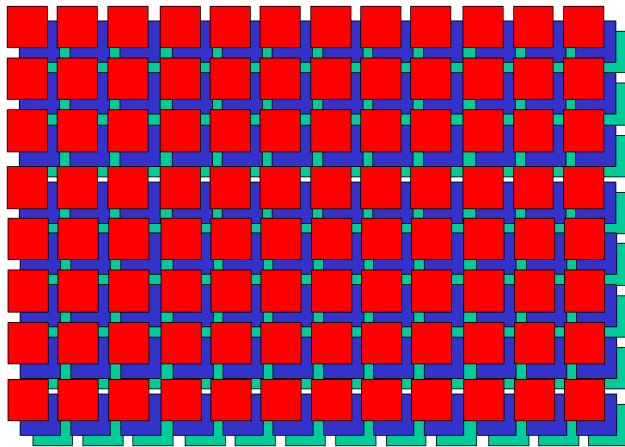


Y

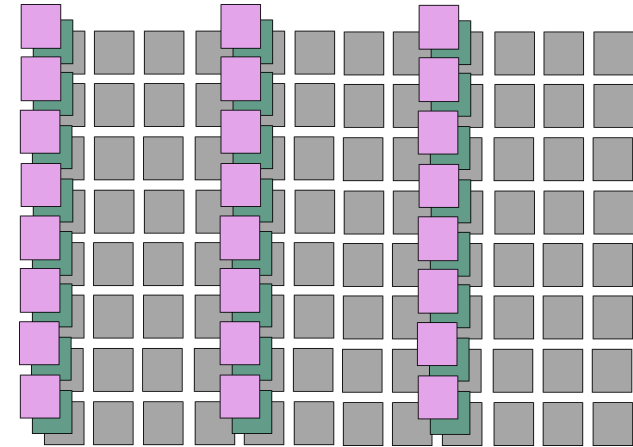
Cb

Cr

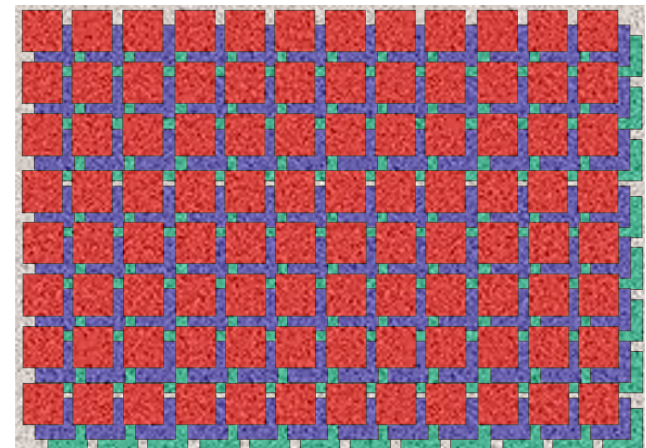
Reception / Decoding



encoding
process



decoding
process



Decoding, reverse process ...
from Chroma sub-sampled to full RGB

- note – interpolating lost information
∴ detail is lost in the process
- Perceptual (e.g. MPEG, etc) encoding +
decoding is also part of this process and loses
a lot more detail

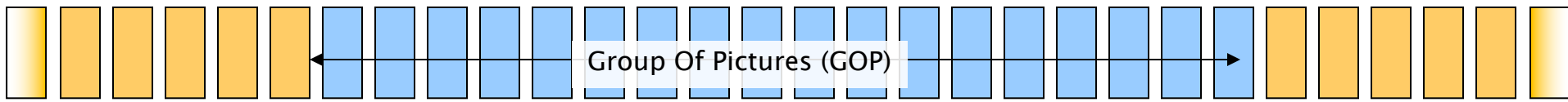
... more later in the module

MPEG – Coding For Moving Images

Intraframe and Interframe Compression

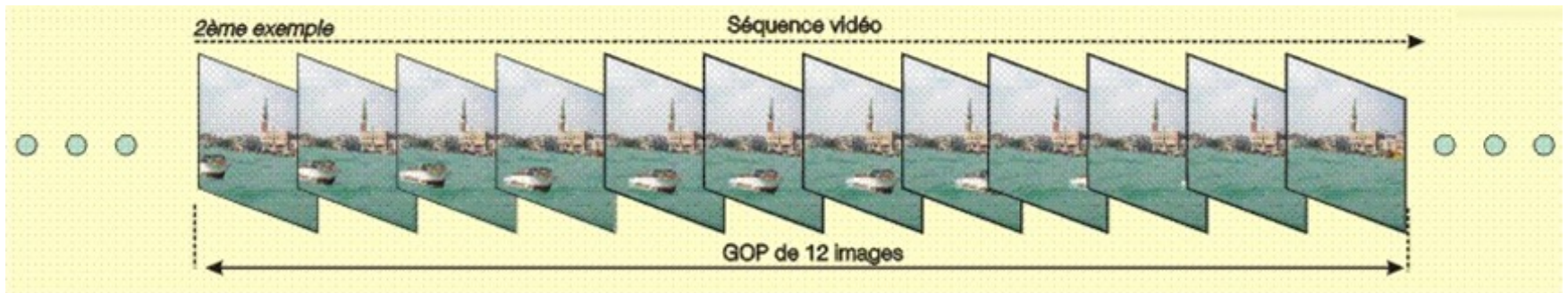
GOPs – Groups Of Pictures

Video Sequence



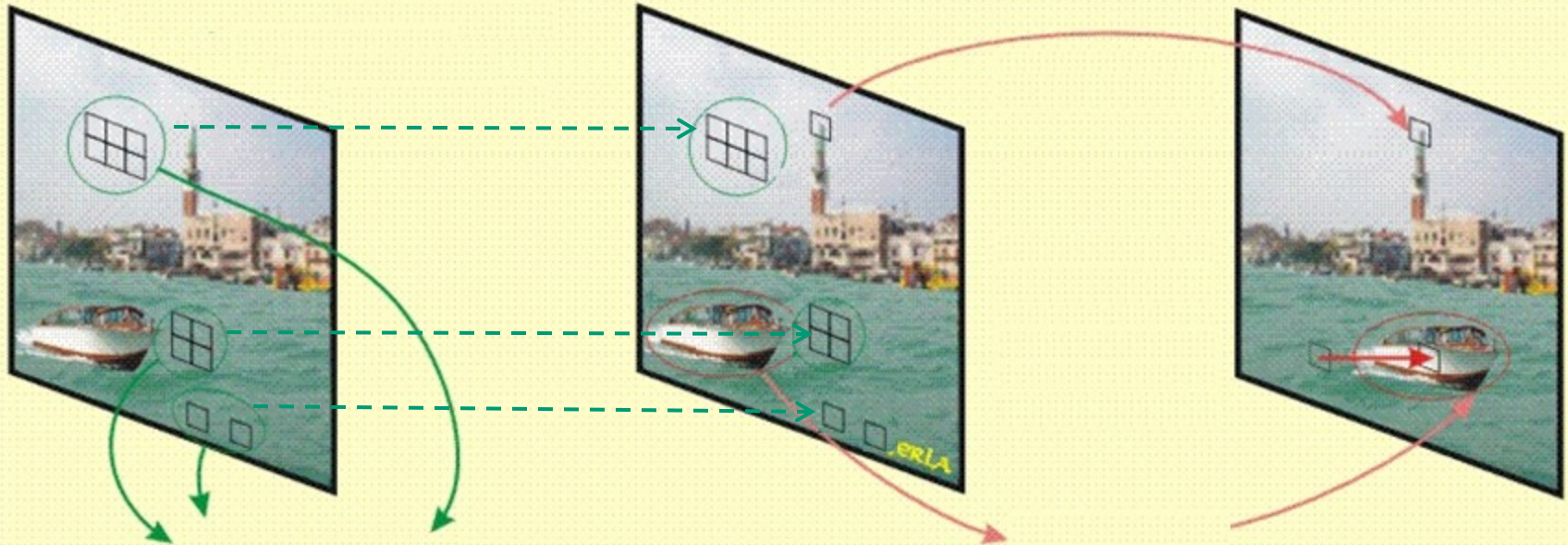
Group of Pictures (GOP)

- Header + series of one or more pictures
- 3 types of pictures: I, P, B
- Provides a regular structure of video content
 - recurring pattern of pictures
 - allow easy access to the video stream



- For TV the GOP is 12 pictures (*0.5 s of a projected at 24 frames / s movie*)
- For NTSC 30 fps, uses 15 pictures

Basic Principles for Moving Elements in Images



Blocks Identical or nearly-identical in following frames

Blocks with *nearly** identical characteristics are replicated; spatial displacement is coded with a *motion vector*

*we'll deal with the "*nearly*" part shortly ...

Difference Coding

Amount of information in the difference between the two frames is a lot less than in Frame 2



Frame 1



Frame 2

Difference:
Frame 2 - 1



- motion in the scene will increase the differences
 - analyse motion
 - ... where each block moved since previous frame
 - encode motion as vector –
 - x- and y- direction vectors
 - encode differences for the moved block like a jpg,
 - i.e. DCT, quantization, RLE (zig-zag), Huffman
- (... but this jpeg is very simple to encode due to little changed information in the image)*

see – <http://www.cs.ucl.ac.uk/teaching/GZ05/08-h261.pdf>

Codecs – Current, Emerging: Tensions in the Market

Codecs – Current, Emerging, Tensions in the Market

New formats – AV1, VVC, EVC, and NDI

- Rising video consumption
- more immersive higher resolution TV
- major new family of video compression standards from
MPEG + ITU-T Study Group 16's VCEG
(Video Coding Experts Group)

Undermining of MPEG HEVC by some of its own patent holders

⇒ traditional codecs versus potentially disruptive alternatives

Cost for distributing video is a major factor

– especially in IP nets (the quality is compressed)

Codecs – Current, Emerging, Tensions in the Market

Current market

- AVC/H.264 codec
 - old, popular, widespread decoder support, well-functioning licensing situation
- HEVC – adoption low in streaming ...

HEVC Advance (licensing admin)

- licensing cost issues
- slow take-up in key market segments
 - ⇒ Rival codecs – better performance / royalty free / or both
- Patent holders had technology accepted before declaring rights and costs in the patent pools
 - further fragmented by HEVC IP holders

Codecs – Current, Emerging, Tensions in the Market

The “Tax on distribution”

- **MPEG LA** – charges per decoding units capped at U\$25m a year
- **HEVC Advance** caps annual unit charge at U\$40m
 - demands royalties for streaming, capped at U\$5m a year
- other HEVC IP owners – Nokia and Technicolor late to join a pool or announce royalty policy
- HEVC royalties significantly more expensive than H.264

Codecs – Current, Emerging, Tensions in the Market

Emerging ... AV1

- Royalty-free (but not yet contested in court ...)
- Based on VP9
 - Alliance for Open Media (AOMedia) (BBC R&D is a member)
- **Few well-conducted comparisons** between AV1 <> HEVC
 - HEVC supporters say HEVC is 30% better than AV1
 - AV1 supporters say AV1 is 40% better than HEVC
 - (see later in these slides ...)
- **AV1 – significantly more complex** encoder to achieve performance
... so not good for real time

Codecs – Current, Emerging, Tensions in the Market

Joint Video Exploration Team (JVET) – VVC Versatile Video Coding (VVC)

MPEG – collaborate with ISO and ITU–VCEG (Video Coding Experts Group):

- formed JVET
 - new suite of video coding technologies
 - basing its work on HEVC
- BBC R&D (Marta Mrak):
 - “HEVC and AV1 are similar; VCC 30% better”
- UHD bitrates similar to current HDTV; storage: 2x as much

Codecs – Current, Emerging, Tensions in the Market

Bitmovin – AV1 will be standard for future content – hi res, VR and AR

Others say – new codec needed

- data demands of 5G – ultra-high bandwidth low latency streaming – VR, 8K – need a new codec – AV1 is not designed for that

Codecs – Current, Emerging, Tensions in the Market

EVC – Essential Video Coding

MPEG-5

Extensible from Royalty free basic profiles

to

licensed more complex profiles

also ...

NewTek

NDI – Royalty Free s/w standard for video device intercommunication

ND1 – Codec – Low latency, ~4:1 UHD

Streaming

- HTTP, HLS
 - MPEG-DASH
- RTP/RTSP

HTTP Streaming

The original solution for streaming

... overtaken by RTP/RTSP based for large IPTV deployments

... more recently back in favour

- used generic HTTP servers (often based on Apache or Windows Server)

Early days

- Limited bandwidth available
 - “Download and play”
 - no *lookahead seeking* or random access
 - Multiple viewers – server bandwidth issues
- Apple introduce progressive download

Streaming Media Protocols – HTTP addresses RTMP Issues

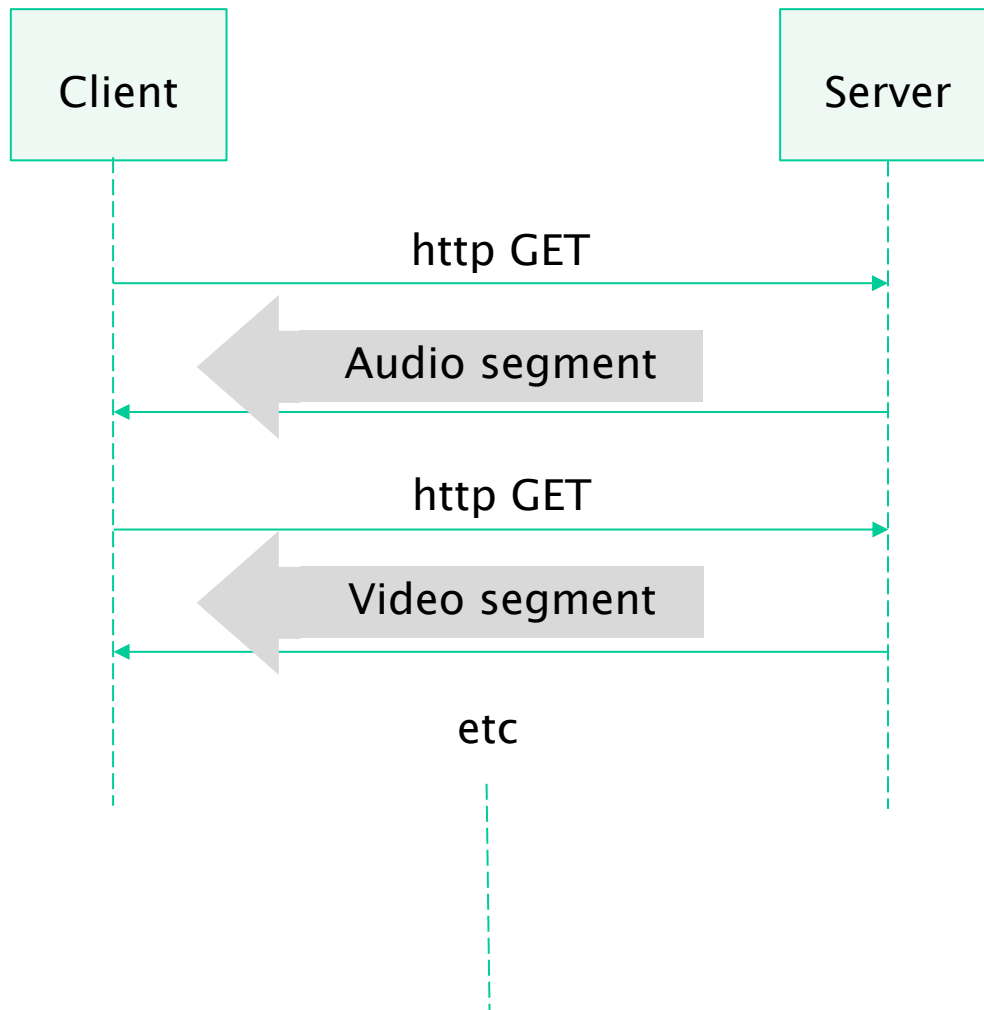
HTTP server/client – *stateless*

- HTTP-streaming: retrieves consecutive short chunks on as-needed basis
 - little waste (the video is delivered as it is watched)
 - server efficiently serves more streams
 - Random seeking
 - (viewer selects – player retrieves the appropriate chunks)
 - firewall friendly
 - can utilise HTTP caching mechanisms
 - less expensive to implement (no streaming server required)
 - scale more cheaply and effectively to serve available users

RTMP/RTSP/HTTP – state of the market

- little impetus for RTMP/RTSP sites to change –still widely used
- new installations – overwhelming view – HTTP
- in fact most video content is delivered via plain old HTTP *progressive download*

HTTP Streaming



Segment timing:

- 10 seconds = reasonable balance for most broadcast content
- Shorter segments = more frequent refreshes of the index file
.. unnecessary network overhead for the client
- Longer segments = latency and initial startup time

Media Formats

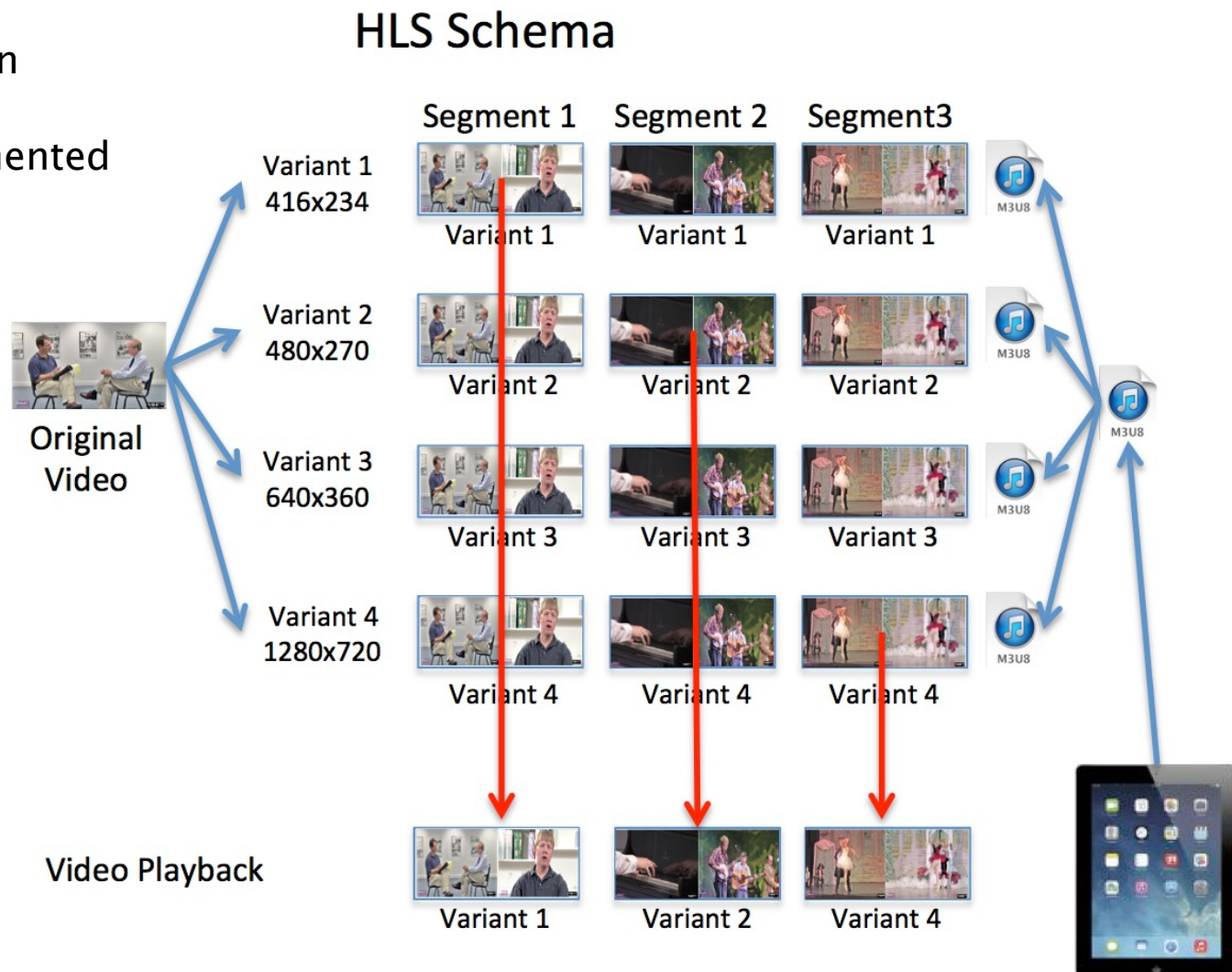
- e.g. video: H.264 / VP9,
audio: .aac / Opus, etc
- video + audio file segments unwrapped
... but certain applications – can be wrapped in MPEG-TS (DVB)

HTTP Live Streaming – HLS

Apple (part of QuickTime
and iOS)

HLS v5 announced in
Sept16

- support for Fragmented MP4



MPEG-DASH

Dynamic Adaptive Streaming over HTTP



Background

- (commercial) audio/video was typically transported over RTP/UDP
 - (but streaming proprietary solutions based HTTP/TCP)
- HTTP streaming – good – provided that the increased latency is acceptable

Benefits:

- **Firewall and NAT traversal** (c.f. services based on RTP/UDP)
- Widely deployed **network caches (CDNs)**
 - Deployment of CDN caches can be kept simple
- Complexity (adaptation) **moved to client**
- Servers are **agnostic** to delivered content
- network resources / configuration may **vary over time**

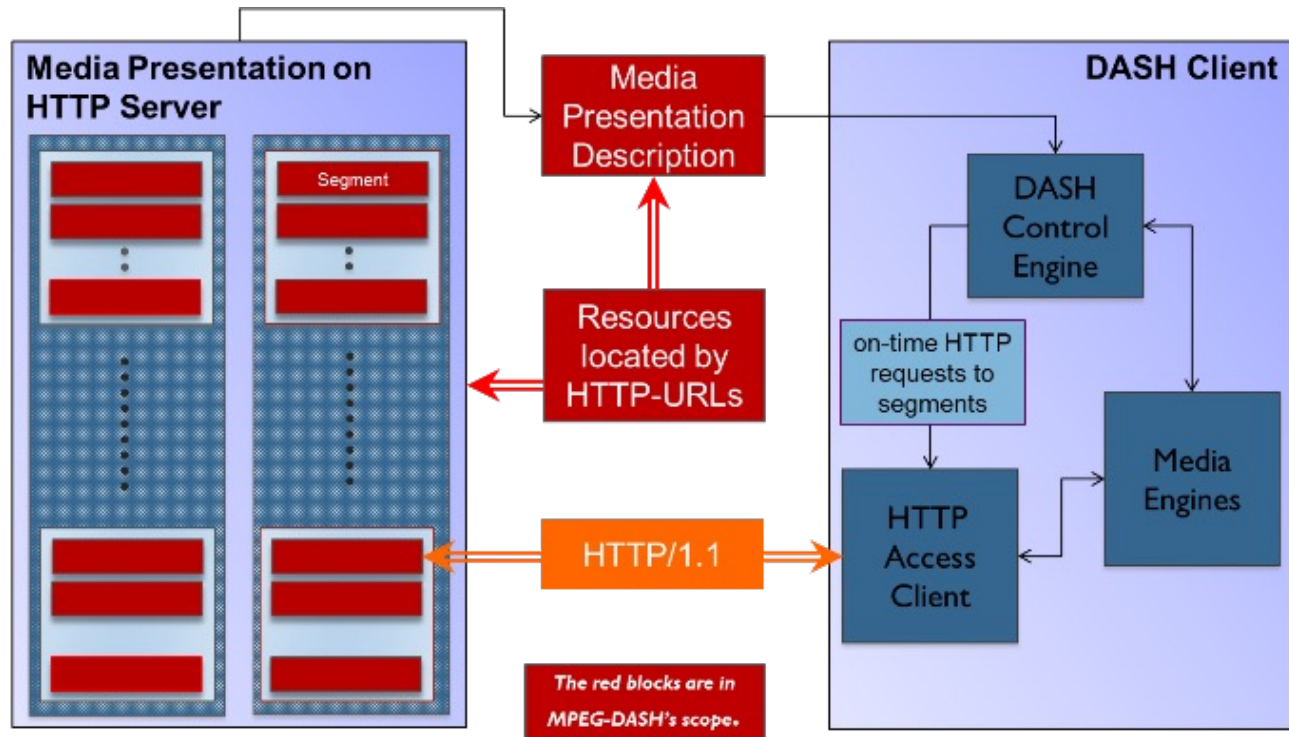
MPEG-DASH

DASH client obtains the MPD ...

MPD (Media Presentation Description) -

XML document

- delivered using HTTP or other transport
- describes media data available at the servers
- allows clients to select the data
- defines segments available
- can either be
 - *initialization segments* - initialization data for decoding
 - *media segments*
- segment format based on MPEG2-TS
 - Using ISO base Media File Format (ISOBMFF - general structure for time-based media
 - (however, segments based on other formats can be specified)





B B C HD

1080 lines